AKTU

Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-0

As per New Syllabus 2022-23

Basic Formulae of Differentiation

Unit-2: Differential Calculus- I

Syllabus

- Successive Differentiation (nth order derivatives)
- Leibnitz theorem
- Partial derivatives
- Euler's Theorem for homogeneous functions
- Total derivative
- Curve tracing

Lec-0

Basic Formulae of Differentiation

① $\frac{d}{dx}(x^n) = n x^{n-1}$

② $\frac{d}{dx}(c) = 0$

③ $\frac{d}{dx}(\sin x) = \cos x$

④ $\frac{d}{dx}(\cos x) = -\sin x$

⑤ $\frac{d}{dx}(\tan x) = \sec^2 x$

⑥ $\frac{d}{dx}(\sec x) = \sec x \tan x$

⑦ $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

⑧ $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

Some Rules of Differentiation

① Product Rule

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

② Quotient Rule

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\left(v\frac{du}{dx} - u\frac{dv}{dx}\right)}{v^2}$$

③ Chain Rule

① $y = \sin(\cos x^2)$

diff wrt x

$$\frac{dy}{dx} = \cos(\cos x^2) \times (-\sin x^2) \times 2x$$

$$\frac{dy}{dx} = -2x\cos(\cos x^2) \times \sin x^2$$

② $y = (2x+3)^4$

diff wrt

$$\frac{dy}{dx} = 4(2x+3)^3 \times 2$$

$$= 8(2x+3)^3$$

① $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$

② $\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$

③ $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

④ $\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$

⑤ $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$

⑥ $\frac{d}{dx}(\operatorname{cosec}^{-1}x) = \frac{-1}{x\sqrt{x^2-1}}$

Differentiation of Exponential and logarithmic function

① $\frac{d}{dx}(a^x) = a^x \log a$

② $\frac{d}{dx}(e^x) = e^x$

③ $\frac{d}{dx}(\log x) = \frac{1}{x}$

① $\sin^2\theta + \cos^2\theta = 1$

② $1 + \tan^2\theta = \sec^2\theta$

③ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

⑤ $\sin 2\theta = 2\sin\theta\cos\theta$

⑥ $\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\cos 2\theta = 1 - 2\sin^2\theta$

$\cos 2\theta = 2\cos^2\theta - 1$

$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$

$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$

⑦ $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

⑧ $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

① $\sin(A+B) = \sin A \cos B + \cos A \sin B$

② $\sin(A-B) = \sin A \cos B - \cos A \sin B$

③ $\cos(A+B) = \cos A \cos B - \sin A \sin B$

④ $\cos(A-B) = \cos A \cos B + \sin A \sin B$

$A > B$

① $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$

② $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

③ $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

④ $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

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Unit-2

Differential Calculus-I

Lec-1

As per New Syllabus 2022-23

Calculation of nth order derivatives

Successive Differentiation (n^{th} order derivatives)

It is differentiation of a function again and again to derive its higher order derivative.

First derivative	Second derivative	Third derivative	n^{th} derivative
$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	$\frac{d^ny}{dx^n}$
y_1	y_2	y_3	y_n
$f'(x)$	$f''(x)$	$f'''(x)$	$f^n(x)$
Dy	D^2y	D^3y	D^ny

Note:

$(n+1)^{th}$ Derivative $= y_{n+1}$

$(n+2)^{th}$ Derivative $= y_{n+2}$

1. n^{th} derivative of $y = x^m$

$$y_1 = m x^{m-1}$$

$$y_2 = m(m-1) x^{m-2}$$

$$y_3 = m(m-1)(m-2) x^{m-3}$$

$$\vdots$$

$$\boxed{y_n = m(m-1)(m-2)(m-3)\cdots\cdots\cdots(m-n+1) x^{m-n}}$$

If $m = n$

$$y_m = m(m-1)(m-2)(m-3)\cdots\cdots 3 \cdot 2 \cdot 1 \cdot x^{m-m}$$

$$y_m = m(m-1)(m-2)(m-3)\cdots\cdots 3 \cdot 2 \cdot 1$$

$$\boxed{y_m = m!}$$

① $y = x^3$

$y_1 = 3x^2$

$y_2 = 3 \times 2 \times x^1$

$y_3 = 3 \times 2 \times 1$

$\boxed{y_3 = 3!}$

$y_4 = 0$

2. n^{th} derivative of $y = (ax+b)^m$, where $n < m$.

$$y = (ax+b)^m$$

$$y_1 = m(ax+b)^{m-1} \times a$$

$$y_2 = m(m-1)(ax+b)^{m-2} \times a^2$$

$$y_3 = m(m-1)(m-2)(ax+b)^{m-3} \times a^3$$

$$\vdots$$

$$\boxed{y_n = m(m-1)(m-2)(m-3)\cdots(m-n+1)(ax+b)^{m-n} \times a^n}$$

If $m = n$

$$y_m = \underbrace{m(m-1)(m-2)\cdots 3 \cdot 2 \cdot 1}(ax+b)^0 a^m$$

$$\boxed{y_m = m!\, a^m}$$

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$$

$\lfloor n$

$\lfloor 5 = 5 \times 4 \times 3 \times 2 \times 1$

3. n^{th} derivative of : $y = \frac{1}{ax+b}$

$$y = \frac{1}{ax+b}$$

$$y = (ax+b)^{-1}$$

$$y_1 = -1(ax+b)^{-2} \times a$$

$$y_2 = -1 \times (-2)(ax+b)^{-3} \times a^2$$

$$y_3 = -1 \times (-2) \times (-3)(ax+b)^{-4} \times a^3$$

$$\vdots$$

$$y_n = -1 \times (-2) \times (-3) \cdots \times (-n)(ax+b)^{-(n+1)} \times a^n$$

$$y_n = (-1)^n \times n! \times (ax+b)^{-(n+1)} a^n$$

$$\boxed{y_n = \frac{(-1)^n \times n! \times a^n}{(ax+b)^{n+1}}}$$

$$y = \frac{1}{x} \qquad (b = 0,\ a = 1)$$

$$\boxed{y_n = \frac{(-1)^n \times n!}{x^{n+1}}}$$

Partial Fraction

TYPE-1

$$\frac{1}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

TYPE-2

$$\frac{1}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{(x-b)}$$

TYPE-3

$$\frac{1}{(ax^2+bx+c)(x-d)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x-d}$$

Q.1.1 If $y=\frac{1}{1-5x+6x^2}$, find y_n

$$y=\frac{1}{1-5n+6n^2}$$

$$y=\frac{1}{6n^2-5n+1}$$

$$y=\frac{1}{(2n-1)(3n-1)}$$

$$\frac{1}{(2n-1)(3n-1)}=\frac{A}{2n-1}+\frac{B}{3n-1}$$

$$\frac{1}{(2n-1)(3n-1)}=\frac{A(3n-1)+B(2n-1)}{(2n-1)(3n-1)}$$

$$1=A(3n-1)+B(2n-1)$$

Put $n=\frac{1}{2}$

$$1=A\left(\frac{3}{2}-1\right)+0$$

$$1=\frac{1}{2}A$$

$$\boxed{A=2}$$

Put $n=\frac{1}{3}$

$$1=0-\frac{1}{3}B$$

$$\boxed{B=-3}$$

$$\frac{1}{(2n-1)(3n-1)}=\frac{2}{(2n-1)}-\frac{3}{(3n-1)}$$

$$y=\frac{2}{2n-1}-\frac{3}{3n-1}$$

$$y_n=2\left[\frac{d^n}{dn^n}\left(\frac{1}{2n-1}\right)\right]-3\left[\frac{d^n}{dn^n}\left(\frac{1}{3n-1}\right)\right]$$

$$y_n=2\left[\frac{(-1)^n\times n!\times(2)^n}{(2n-1)^{n+1}}\right]-3\left[\frac{(-1)^n\times n!\times(3)^n}{(3n-1)^{n+1}}\right]$$

$$\boxed{y_n=\frac{(-1)^n\times n!\times(2)^{n+1}}{(2n-1)^{n+1}}-\frac{(-1)^n\times n!\times(3)^{n+1}}{(3n-1)^{n+1}}}$$

Q.1.2 *Find the nth derivative of* $\dfrac{2x+1}{(2x+1)(2x+3)}$

Let $y = \dfrac{2x+1}{(2x-1)(2x+3)}$

$$\frac{2x+1}{(2x-1)(2x+3)} = \frac{A}{2x-1} + \frac{B}{2x+3}$$

$$2x+1 = A(2x+3) + B(2x-1)$$

Put $x = \frac{1}{2}$

$$2\times\frac{1}{2}+1 = A\left(2\times\frac{1}{2}+3\right) + 0$$

$$2 = 4A$$

$$A = \frac{2}{4}$$

$$A = \frac{1}{2}$$

Put $x = -\frac{3}{2}$

$$2\times\left(-\frac{3}{2}\right)+1 = 0 + B\left[2\times\left(-\frac{3}{2}\right)-1\right]$$

$$-3+1 = 0 - 4B$$

$$+2 = +4B$$

$$B = \frac{1}{2}$$

$$\frac{2x+1}{(2x-1)(2x+3)} = \frac{1}{2(2x-1)} + \frac{1}{2(2x+3)}$$

$$y = \frac{1}{2(2x-1)} + \frac{1}{2(2x+3)}$$

$$y_n = \frac{1}{2}\left[\frac{d^n}{dx^n}\left(\frac{1}{2x-1}\right)\right] + \frac{1}{2}\left[\frac{d^n}{dx^n}\left(\frac{1}{2x+3}\right)\right]$$

$$y_n = \frac{1}{2}\left[\frac{(-1)^n \times n! \times 2^n}{(2x-1)^{n+1}}\right] + \frac{1}{2}\left[\frac{(-1)^n \times n! \times 2^n}{(2x+3)^{n+1}}\right]$$

$$y_n = \frac{1}{2}(-1)^n \times n! \times 2^n\left[\frac{1}{(2x-1)^{n+1}} + \frac{1}{(2x+3)^{n+1}}\right]$$

$$y = (-1)^n \times n! \times 2^{n-1}\left[\frac{1}{(2x-1)^{n+1}} + \frac{1}{(2x+3)^{n+1}}\right]$$

Practice Q.1.1 If $y=\frac{5x+12}{6+5x+x^2}$, find y_n

4. n^{th} derivative of $y = \log(ax+b)$

$$y = \log(ax+b)$$

$$y_1 = \frac{1}{ax+b} \times a$$

$$y_1 = (ax+b)^{-1} \times a$$

$$y_2 = -1(ax+b)^{-2} \times a^2$$

$$y_3 = -1 \times (-2) \times (ax+b)^{-3} \times a^3$$

$$\vdots$$

$$y_n = (-1)^{n-1} \times (n-1)! (ax+b)^{-n} \times a^n$$

$$\boxed{y_n = \frac{(-1)^{n-1} \times (n-1)! \times a^n}{(ax+b)^n}}$$

If $y = \log x \quad (b=0,\ a=1)$

$$\boxed{y_n = \frac{(-1)^{n-1} \times (n-1)!}{x^n}}$$

Q.1.3 *Find the nth* derivative of $\log(ax + x^2)$

$$y = \log(ax + x^2)$$

$$y = \log x(a + x)$$

$$y = \log x(x + a)$$

$$y = \log x + \log(x + a)$$

$$\{\because \log mn = \log m + \log n\}$$

$$y_n = \frac{d^n}{dx^n}(\log x) + \frac{d^n}{dx^n}\{\log(x+a)\}$$

$$y_n = \frac{(-1)^{n-1} \times (n-1)!}{x^n} + \frac{(-1)^{n-1} \times (n-1)!}{(x+a)^n}$$

$$\boxed{y_n = (-1)^{n-1} \times (n-1)! \left[\frac{1}{x^n} + \frac{1}{(x+a)^n}\right]}$$

Practice Q.1.2 $Find\ the\ n^{th}$ derivative of $\log(3x)$


Thank
you

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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-2

As per New Syllabus 2022-23

Calculation of nth order derivatives - 2

1. n^{th} derivative of $y = x^m$

2. n^{th} derivative of $y = (ax+b)^m$, where $n < m$.

3. n^{th} derivative of $y = \frac{1}{ax+b}$

4. n^{th} derivative of $y = \log(ax+b)$

Partial Fraction

5. n^{th} derivative of exponential function

(i) $y = a^{mx}$

$$y_1 = m\,a^{mx}\log a$$

$$y_2 = m^2\,a^{mx}(\log a)^2$$

$$y_3 = m^3\,a^{mx}(\log a)^3$$

$$\vdots$$

$$\boxed{y_n = m^n\,a^{mx}\times(\log a)^n}$$

$y = a^x$

$$y_n = (1)^n\,a^x(\log a)^n$$

$$\boxed{y_n = a^x(\log a)^n}$$

(ii) $y = e^{mx}$

$$y_n = m^n\,e^{mx}\times(\log e)^n$$

$$\boxed{y_n = m^n\,e^{mx}}$$

$$\boxed{\log e = 1}$$

6. n^{th} derivative of $y = \sin(ax + b)$

$y = \sin(ax+b)$

$y_1 = a\cos(ax+b) =$

$$y_1 = a\sin\left(ax+b+\frac{\pi}{2}\right)$$

$y_2 = a^2\cos\left(ax+b+\frac{\pi}{2}\right)$

$$y_2 = a^2\sin\left(ax+b+2\frac{\pi}{2}\right)$$

$y_3 = a^3\cos\left(ax+b+2\frac{\pi}{2}\right)$

$y_3 = a^3\sin\left(ax+b+3\frac{\pi}{2}\right)$

$y_n = a^n\sin\left(ax+b+n\frac{\pi}{2}\right)$

$$y_n = a^n\sin\left(ax+b+n\frac{\pi}{2}\right)$$

$y = \sin ax$

$$y_n = a^n\sin\left(ax+n\frac{\pi}{2}\right)$$

$y = \sin x$

$y_n = (1)^n\sin\left(x+n\frac{\pi}{2}\right)$

$$y_n = \sin\left(x+n\frac{\pi}{2}\right)$$

7. n^{th} derivative of $y = \cos(ax + b)$

$$y_n = a^n\cos\left(ax+b+n\frac{\pi}{2}\right)$$

$y = \cos ax$

$$y_n = a^n\cos\left(ax+n\frac{\pi}{2}\right)$$

$y = \cos x$

$y_n = (1)^n\cos\left(x+n\frac{\pi}{2}\right)$

$$y_n = \cos\left(x+n\frac{\pi}{2}\right)$$

Q.2.1 Find the nth derivative of $y = \sin^3 x$

$$y = \sin^3 x$$

$$y = \frac{1}{4}\left[3\sin x - \sin 3x\right]$$

$$y_n = \frac{1}{4}\left[3\frac{d^n}{dx^n}(\sin x) - \frac{d^n}{dx^n}(\sin 3x)\right]$$

$$y_n = \frac{1}{4}\left[3\sin\left(x + n\frac{\pi}{2}\right) - 3^n \sin\left(3x + n\frac{\pi}{2}\right)\right]$$

$$\boxed{y_n = \frac{3}{4}\sin\left(x + n\frac{\pi}{2}\right) - \frac{3^n}{4}\sin\left(3x + n\frac{\pi}{2}\right)}$$

Formulae

① $\sin 3x = 3\sin x - 4\sin^3 x$

$4\sin^3 x = 3\sin x - \sin 3x$

$\sin^3 x = \frac{1}{4}\left(3\sin x - \sin 3x\right)$

② $\frac{d^n}{dx^n}(\sin x) = \sin\left(x + n\frac{\pi}{2}\right)$

③ $\frac{d^n}{dx^n}(\sin 3x) = 3^n \sin\left(3x + n\frac{\pi}{2}\right)$

Q.2.2 Find the nth derivative of $y = \sin x . \sin 2x$

$$y = \sin x \cdot \sin 2x$$

$$y = \frac{1}{2}(2 \sin x \cdot \sin 2x)$$

$$y = \frac{1}{2}[\cos(x-2x) - \cos(x+2x)]$$

$$y = \frac{1}{2}[\cos(-x) - \cos 3x]$$

$$y = \frac{1}{2}[\cos x - \cos 3x]$$

$$y_n = \frac{1}{2}\left[\frac{d^n}{dx^n}(\cos x) - \frac{d^n}{dx^n}(\cos 3x)\right]$$

$$y_n = \frac{1}{2}\left[\cos\left(x + n\frac{\pi}{2}\right) - 3^n \cos\left(3x + n\frac{\pi}{2}\right)\right]$$

$$\boxed{y_n = \frac{1}{2}\cos\left(x + n\frac{\pi}{2}\right) - \frac{3^n}{2}\cos\left(3x + n\frac{\pi}{2}\right)}$$

Formulae

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\cos(-x) = \cos x$$

Q.2.3 Find the nth derivative of $y = \sin^2 x . \cos^3 x$

$$y = \sin^2 x \cdot \cos^3 x$$

$$y = (\sin^2 x \times \cos^2 x) \times \cos x$$

$$y = (\sin x \cos x)^2 \times \cos x$$

$$y = \left[\frac{1}{2}(2\sin x \cos x)\right]^2 \cos x$$

$$y = \frac{1}{4}(\sin 2x)^2 \cos x$$

$$y = \frac{1}{4}\left[\frac{1-\cos 4x}{2}\right]\cos x$$

$$y = \frac{1}{8}\left[\cos x - 2\cos 4x \cos x\right]$$

$$y = \frac{1}{2\times 8}\left[2\cos x - \{\cos(4x+x) + \cos(4x-x)\}\right]$$

$$y = \frac{1}{2\times 8}\left[2\cos x - \cos 5x - \cos 3x\right]$$

$$y_n = \frac{1}{2\times 8}\left[\frac{d^n}{dx^n}(2\cos x) - \frac{d^n}{dx^n}(\cos 5x) - \frac{d^n}{dx^n}(\cos 3x)\right]$$

$$y_n = \frac{1}{16}\left[2\cos\left(x + n\frac{\pi}{2}\right) - 5^n \cos\left(5x + n\frac{\pi}{2}\right) - 3^n \cos\left(3x + n\frac{\pi}{2}\right)\right]$$

8. n^{th} derivative of $y = e^{ax}\sin(bx+c)$

$$y = e^{ax}\sin(bx+c)$$

$$y_1 = e^{ax}\cos(bx+c)\times b + \sin(bx+c)\times e^{ax}\times a$$

$$y_1 = e^{ax}\left[b\cos(bx+c) + a\sin(bx+c)\right]$$

Put $a = r\cos\theta$ — ①

$b = r\sin\theta$ — ②

$$a^2+b^2 = r^2\cos^2\theta + r^2\sin^2\theta$$

$$a^2+b^2 = r^2(\sin^2\theta+\cos^2\theta)$$

$$\boxed{r = \sqrt{a^2+b^2}}$$

$$\tan\theta = \frac{b}{a}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{b}{a}\right)}$$

9. n^{th} derivative of $y = e^{ax}\cos(bx+c)$

$$\boxed{y_n = r^n e^{ax}\cos(bx+c+n\theta)}$$

$$y_1 = e^{ax}\left[r\sin\theta\cos(bx+c) + r\cos\theta\sin(bx+c)\right]$$

$$y_1 = e^{ax} r\left[\sin\theta\cos(bx+c) + \cos\theta\sin(bx+c)\right]$$

$$y_1 = e^{ax} r\left[\sin(bx+c+\theta)\right]$$

$$\boxed{y_n = e^{ax} r^n \sin(bx+c+n\theta)}$$

Q.2.4 Find the nth derivative of $y = e^x . \sin^3 x$

$$y = e^x \sin^3 x$$

$$y = e^x \times \frac{1}{4}\left[3\sin x - \sin 3x\right]$$

$$y = \frac{1}{4}\left[3e^x \sin x - e^x \sin 3x\right]$$

$$y = \frac{3}{4}e^x \sin x - \frac{1}{4}e^x \sin 3x$$

$$y_n = \frac{3}{4}\frac{d^n}{dx^n}(e^x \sin x) - \frac{1}{4}\frac{d^n}{dx^n}(e^x \sin 3x)$$

$$y_n = \frac{3}{4}r^n e^x \sin(x + n\theta) - \frac{1}{4}r^n e^x \sin(3x + n\theta)$$

$$r = \sqrt{a^2+b^2} = \sqrt{1+3^2} = \sqrt{10} = (10)^{\frac{1}{2}}$$

$$\theta = \tan^{-1}\frac{b}{a} = \tan^{-1} 3$$

$$r = \sqrt{1^2+1^2} = \sqrt{2} = (2)^{\frac{1}{2}}$$

$$\theta = \tan^{-1}\frac{b}{a} = \tan^{-1}\frac{1}{1} = \frac{\pi}{4}$$

$$y_n = \frac{3}{4}(2)^{\frac{n}{2}} e^x \sin\left(x + n\frac{\pi}{4}\right) - \frac{1}{4}(10)^{\frac{n}{2}} e^x \sin(3x + n\tan^{-1}(3))$$

$$y_n = \frac{3}{4}(2)^{\frac{n}{2}} e^x \sin\left(x + n\frac{\pi}{4}\right) - \frac{1}{4}(10)^{\frac{n}{2}} e^x \sin(3x + n\tan^{-1} 3)$$

Q.2.5 Find the nth derivative of $y = e^x . \sin x . \cos x$

$$y = e^x \sin x \cos x$$

$$y = e^x \frac{2 \sin x \cos x}{2}$$

$$y = \frac{1}{2} e^x \sin 2x$$

$$a = 1, \quad b = 2, \quad c = 0$$

$$r = \sqrt{1^2 + 2^2} = \sqrt{5} = (5)^{\frac{1}{2}}$$

$$\theta = \tan^{-1} \frac{b}{a} = \tan^{-1} \frac{2}{1}$$

$$\boxed{\theta = \tan^{-1} 2}$$

$$y_n = \frac{1}{2} \frac{d^n}{dx^n} \left(e^x \sin 2x \right)$$

$$y_n = \frac{1}{2} r^n e^{ax} \sin\left(bx + c + n\theta \right)$$

$$\boxed{y_n = \frac{1}{2} (5)^{\frac{n}{2}} e^x \sin\left(2x + n \tan^{-1} 2 \right)}$$


Thank
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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-3

As per New Syllabus 2022-23

Leibnitz Theorem Part-1

Leibnitz Theorem

If $y = uv$

$$y_n = \left({}^nC_0\, u_n v\right) + \left({}^nC_1\, u_{n-1} v_1\right) + \left({}^nC_2\, u_{n-2} v_2\right) + \text{- - - - - -} + \left({}^nC_n\, u v_n\right)$$

where $\boxed{{}^nC_r = \dfrac{n!}{r!\,(n-r)!}}$

$${}^nC_0 = \frac{n!}{0!\,(n-0)!} = \frac{n!}{1 \times n!} = 1$$

$${}^nC_0 = 1$$

$${}^nC_1 = n$$

$${}^nC_2 = \frac{n(n-1)}{2}$$

$${}^nC_n = 1$$

Q.3.1 If $y = e^{\tan^{-1}x}$, prove that $(1+x^2)y_{n+2} + [(2n+2)x-1]y_{n+1} + n(n+1)y_n = 0$

$y = e^{\tan^{-1}x}$

$y_1 = e^{\tan^{-1}x} \times \frac{1}{1+x^2}$

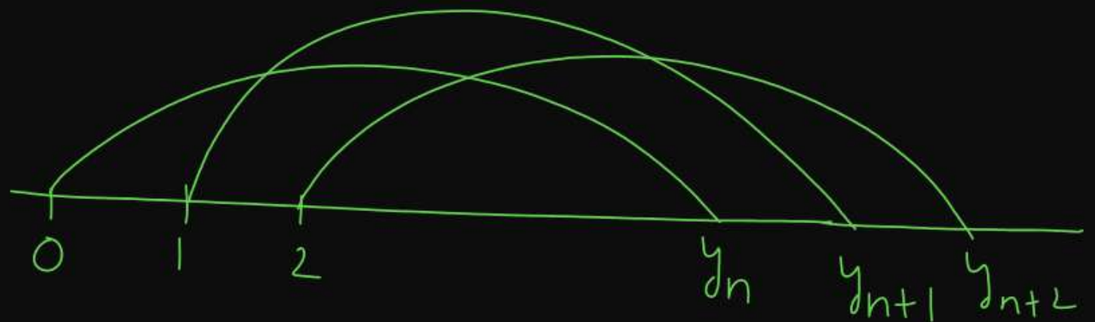
$(1+x^2)y_1 = e^{\tan^{-1}x}$

$(1+x^2)y_1 = y$

$(1+x^2)y_2 + y_1 \times 2x = y_1$

$(1+x^2)y_2 + 2xy_1 - y_1 = 0$

$(1+x^2)y_2 + (2x-1)y_1 = 0$



$$(1+x^2)y_2 + (2x-1)y_1 = 0$$

Apply Leibnitz theorem

$$n_{C_0} \times y_{n+2}(1+x^2) + n_{C_1} y_{n+1} \times 2x + n_{C_2} y_n \times 2 + n_{C_0} y_{n+1}(2x-1) + n_{C_1} y_n \times 2 = 0$$

$$(1+x^2)y_{n+2} + 2nx y_{n+1} + \frac{n(n-1)}{2} \times y_n \times 2 + (2x-1)y_{n+1} + 2n y_n = 0$$

$$(1+x^2)y_{n+2} + 2nx y_{n+1} + n(n-1)y_n + (2x-1)y_{n+1} + 2n y_n = 0$$

$$(1+x^2)y_{n+2} + [2nx + 2x - 1]y_{n+1} + [n(n-1) + 2n]y_n = 0$$

$$\boxed{(1+x^2)y_{n+2} + [(2n+2)x - 1]y_{n+1} + n(n+1)y_n = 0}$$

Hence proved

$$n(n-1) + 2n$$
$$= n^2 - n + 2n$$
$$= n^2 + n$$
$$= n(n+1)$$

Q.3.2 *If* $y = e^{m \sin^{-1} x}$, *show that* $(1-x^2)\,y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$

$$y = e^{m\sin^{-1}x}$$

$$y_1 = e^{m\sin^{-1}x} \times \frac{m}{\sqrt{1-x^2}}$$

$$y_1 = \frac{my}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2}\, y_1 = my$$

squaring both sides

$$(1-x^2)\, y_1^2 = m^2 y^2$$

Again diff. w.r.t x

$$(1-x^2)\,2y_1y_2 + y_1^2(-2x) = m^2 \times 2yy_1$$

$$(1-x^2)\,2y_1y_2 - 2xy_1^2 = 2m^2yy_1$$

$$2y_1\left[(1-x^2)y_2 - xy_1\right] = 2y_1(m^2y)$$

$$(1-x^2)y_2 - xy_1 = m^2y$$

$$(1-x^2)y_2 - xy_1 = m^2 y$$

Apply Leibnitz theorem

$$\binom{n}{0} y_{n+2}(1-x^2) + \binom{n}{1} y_{n+1}(-2x) + \binom{n}{2} y_n(-2) - \left[\binom{n}{0} y_{n+1} x^n + \binom{n}{1} y_n x\right] = m^2 y_n$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2} y_n \cdot 2 - ny_{n+1} - nxy_n = m^2 y_n$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n(n-1)y_n - ny_n - m^2 y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - [n(n-1)+n+m^2]y_n = 0$$

$$n(n-1)+n+m^2$$
$$n^2 - n + n + m^2$$
$$(n^2+m^2)$$

$$\boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0}$$

Q.3.3 *If* $y = x^n \log x$ *, prove that* $y_{n+1} = \frac{n!}{x}$

$y = x^n \log x$

diff. wrt to x

$y_1 = x^n \times \frac{1}{x} + \log x \times n x^{n-1}$

$y_1 = \frac{x^n + \log x \times n x^{n-1} \times x}{x}$

$y_1 = \frac{x^n + n x^n \log x}{x}$

$x y_1 = x^n + n x^n \log x$

$x y_1 = n^n + n y$

Apply Leibnitz theorem

$n_{C_0} y_{n+1} x + n_{C_1} y_n \times 1 = n! + n y_n$

$x y_{n+1} + n y_n = n! + n y_n$

$x y_{n+1} = n!$

$$y_{n+1} = \frac{n!}{x}$$

Hence proved

Q.3.4 $If\ y = \sin(m\ sin^{-1}x)$ Prove that $(1-x^2)\,y_{n+2} - (2n+1)xy_{n+1} - (n^2-m^2)y_n = 0$

Q.3.5 If $y = a\cos(\log x) + b\sin(\log x)$,

prove that $x^2 y_{n+2} + (2n+1)x\, y_{n+1} + (n^2+1)y_n = 0$


Thank
you

AKTU

Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-4

As per New Syllabus 2022-23

Leibnitz Theorem Part-2

Value of the nth derivative of a function for x=0

Working rule to find the value of nth derivative of a function for x = 0

Step 1. Equate the given function to y. (1)

Step 2. Find y_1. (2)

Step 3. Again find y_2. (3)

Step 4. Differentiate both sides n-times by Leibnitz Theorem. (4)

Step 5. Put x = 0 in equations of (1), (2), (3) and (4). (5)

Step 6. On putting x = 0, two cases arises (i) when n = odd (ii) when n = even integer

Q.4.1 If $y = \sin(a \sin^{-1}x)$, find $y_n(0)$.

$y = \sin(a\sin^{-1}x)$ — ①

$y_1 = \cos(a\sin^{-1}x) \times \frac{a}{\sqrt{1-x^2}}$ — ②

$(\sqrt{1-x^2})y_1 = a\cos(a\sin^{-1}x)$

squaring both side

$(1-x^2)y_1^2 = a^2\cos^2(a\sin^{-1}x)$

$(1-x^2)y_1^2 = a^2[1-\sin^2(a\sin^{-1}x)]$

$(1-x^2)y_1^2 = a^2(1-y^2)$

$(1-x^2)2y_1y_2 + y_1^2(-2x) = a^2(-2yy_1)$

$2y_1[(1-x^2)y_2 - xy_1] = 2y_1(-a^2y)$

$(1-x^2)y_2 - xy_1 + a^2y = 0$ — ③

Apply Leibnitz theorem

$nC_0\, y_{n+2}(1-x^2) + nC_1\, y_{n+1}(-2x) + nC_2\, y_n(-2) - \left[nC_0\, y_{n+1}x + nC_1\, y_n \times 1\right] + a^2y_n = 0$

$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2}y_n \times 2 - ny_{n+1} - ny_n + a^2y_n = 0$

$(1-x^2)y_{n+2} - 2nxy_{n+1} - xy_{n+1} - n^2y_n + ny_n - ny_n + a^2y_n = 0$

$\boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-a^2)y_n = 0}$ — ④

Put $n=0$ in (1) (2) (3) and (4)

$y(0) = 0$

$y_1(0) = a$

$y_2(0) = -a^2 y(0) = 0$

$$y_{n+2}(0) = (n^2 - a^2) y_n(0) \quad \text{—— (5)}$$

Put $n = 1, 2, 3, 4 \ldots$ in (5)

$$y_3(0) = (1^2 - a^2) y_1(0) = (1^2 - a^2) a$$

$$y_4(0) = (2^2 - a^2) y_2(0) = 0$$

$$y_5(0) = (3^2 - a^2) y_3(0) = (3^2 - a^2)(1^2 - a^2) a$$

$$y_6(0) = (4^2 - a^2) y_4(0) = 0$$

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$$y_n = \begin{cases} 0 & \text{when } n \text{ is even} \\ [(n-2)^2 - a^2][(n-4)^2 - a^2] \ldots (3^2 - a^2)(1^2 - a^2) a & \text{when } n \text{ is odd} \end{cases}$$

Q.4.2 *If* $y = [\sin^{-1}x]^2$, *prove that* $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

Also find the nth derivative of y for $x=0$.

$y = (\sin^{-1}x)^2$ — ①

$y_1 = 2(\sin^{-1}x) \times \frac{1}{\sqrt{1-x^2}}$ — ②

$\sqrt{1-x^2}\,y_1 = 2\sin^{-1}x$

squaring both side

$(1-x^2)y_1^2 = 4(\sin^{-1}x)^2$

$(1-x^2)y_1^2 = 4y$

$(1-x^2)2y_1y_2 + y_1^2 \times (-2x) = 4y_1$

$2y_1[(1-x^2)y_2 - xy_1 = 2y_1(2)$

$(1-x^2)y_2 - xy_1 = 2$

$(1-x^2)y_2 - xy_1 - 2 = 0$ — ③

Apply Leibnitz theorem

$n_0 y_{n+2}(1-x^2) + n_1 y_{n+1}(-2x) + n_2 y_n(-2) - [n_0 y_{n+1} \times x + n_1 y_n \times 1] = 0$

$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2} y_n \times 2 - xy_{n+1} - ny_n = 0$

$(1-x^2)y_{n+2} - 2nxy_{n+1} - xy_{n+1} - n^2y_n + ny_n - ny_n = 0$

$$\boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0}$$ — ④

Put $n = 0$ in (1) (2) (3) and (4)

$y(0) = 0$

$y_1(0) = 0$

$y_2(0) = 2$

$$y_{n+2}(0) = n^2 y_n(0) \quad \text{— (5)}$$

Put $n = 1, 2, 3, 4 \ldots$ in (5)

$$y_3(0) = 1^2 y_1(0) = 0$$

$$y_4(0) = 2^2 y_2(0) = 2^2 \cdot 2$$

$$y_5(0) = 3^2 y_3(0) = 0$$

$$y_6(0) = 4^2 y_4(0) = 4^2 \cdot 2^2 \cdot 2$$

$$y_n = \begin{cases} 0, \text{ when } n \text{ is odd} \\ (n-2)^2 \cdot (n-4)^2 \cdots 4^2 \cdot 2^2 \cdot 2 \\ \text{when } n \text{ is even} \end{cases}$$

Q.4.3 *If* $y = e^{m\cos^{-1}x}$, *show that* $(1-x^2)\,y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$

$$y = e^{m\cos^{-1}x} \quad -(1)$$

$$y_1 = e^{m\cos^{-1}x} \times \frac{m(-1)}{\sqrt{1-x^2}} \quad -(2)$$

$$(\sqrt{1-x^2})y_1 = -m e^{m\cos^{-1}x}$$

$$(\sqrt{1-x^2})y_1 = -my$$

Squaring both side

$$(1-x^2)y_1^2 = m^2y^2$$

$$(1-x^2)2y_1y_2 + y_1^2(-2x) = m^2 2yy_1$$

$$2y_1[(1-x^2)y_2 - xy_1] = 2y_1(m^2y)$$

$$(1-x^2)y_2 - xy_1 - m^2y = 0 \quad -(3)$$

Apply Leibnitz theorem

$$n_{C_0}\,y_{n+2}(1-x^2) + n_{C_1}\,y_{n+1}(-2x) + n_{C_2}\,y_n(-2) - \left[n_{C_0}\,y_{n+1}\,x + n_{C_1}\,y_n \times 1\right] - m^2y_n = 0$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2}y_n \times 2 - xy_{n+1} - ny_n - m^2y_n = 0$$

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - xy_{n+1} - n^2y_n + n\cancel{y_n} - n\cancel{y_n} - m^2y_n = 0$$

$$\boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0} \quad -(4)$$

Put $x = 0$ in (1) (2) (3) (4)

Calculate $y_n(0)$

$$y(0) = e^{\frac{m\pi}{2}}$$

$$y_1(0) = -m e^{\frac{m\pi}{2}}$$

$$y_2(0) = m^2 y(0) = m^2 e^{\frac{m\pi}{2}}$$

$$\boxed{y_{n+2}(0) = (n^2+m^2) y_n(0)} \quad \text{— (5)}$$

Put $n = 1, 2, 3, 4 \ldots$ in (5)

$$y_3(0) = (1^2+m^2) y_1(0) = -m(1^2+m^2) e^{\frac{m\pi}{2}}$$

$$y_4(0) = (2^2+m^2) y_2(0) = m^2(2^2+m^2) e^{\frac{m\pi}{2}}$$

$$y_5(0) = (3^2+m^2) y_3(0) = -m(1^2+m^2)(3^2+m^2) e^{\frac{m\pi}{2}}$$

$$y_6(0) = (4^2+m^2) y_4(0) = m^2(2^2+m^2)(4^2+m^2) e^{\frac{m\pi}{2}}$$

$$y_n = \begin{cases} -m(1^2+m^2)(3^2+m^2) \ldots [(n-2)^2+m^2] e^{\frac{m\pi}{2}} & \text{when } n \text{ is odd} \\ m^2(2^2+m^2)(4^2+m^2) \ldots [(n-2)^2+m^2] e^{\frac{m\pi}{2}} & \text{when } n \text{ is even} \end{cases}$$

Q.4.4 If $y = [x + \sqrt{1 + x^2}]^m$, find $y_n(0)$

$$y = [x + \sqrt{1+x^2}]^m \quad \text{—— ①}$$

Taking log both side

$$\log y = \log[x + \sqrt{1+x^2}]^m$$

$$\log y = m \log[x + \sqrt{1+x^2}]$$

diff wrt x

$$\frac{1}{y} y_1 = m \times \frac{1}{(x + \sqrt{1+x^2})} \times \left(1 + \frac{1}{2\sqrt{1+x^2}} \times 2x\right)$$

$$\frac{1}{y} y_1 = m \times \frac{1}{(x + \sqrt{1+x^2})} \left[\frac{(\sqrt{1+x^2} + x)}{\sqrt{1+x^2}}\right]$$

$$\frac{1}{y} y_1 = \frac{m}{\sqrt{1+x^2}}$$

$$y_1 = \frac{my}{\sqrt{1+x^2}} \quad \text{— ②}$$

$$(\sqrt{1+x^2}) y_1 = my$$

squaring both side

$$(1+x^2) y_1^2 = m^2 y^2$$

$$(1+x^2) 2 y_1 y_2 + y_1^2 \times 2x = m^2 \times 2 y y_1$$

$$2y_1 [(1+x^2) y_2 + x y_1] = 2 y_1 (m^2 y)$$

$$\boxed{(1+x^2) y_2 + x y_1 - m^2 y = 0} \quad \text{— ③}$$


Thank
you

AKTU

Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-5

As per New Syllabus 2022-23

Partial Derivative of first order

Part-1

Partial Derivative

1. Partial Derivative of first order
2. Partial Derivative second order
3. Euler's Theorem for homogenous functions
4. Total Derivative

Partial Derivative of first order

Let z = f (x,y)

The derivative of z with respect to x , keeping y as constant is called partial derivative of z w.r.t. to x.

$\frac{\partial z}{\partial x}$, $\frac{\partial f}{\partial x}$, $f_x(x,y)$, f_x etc

Similarly partial derivative of z w.r.t. to y keeping x as constant.

$\frac{\partial z}{\partial y}$, $\frac{\partial f}{\partial y}$, $f_y(x,y)$, f_y etc

Q.5.1 $V = x^4y$, *find* (i) $\frac{\partial V}{\partial x}$ (ii) $\frac{\partial V}{\partial y}$

$V = x^4 y$

(i) $\frac{\partial V}{\partial x} = 4x^3 y$

(ii) $\frac{\partial V}{\partial y} = x^4$

(i) $\frac{d}{dx}(x^n) = n x^{n-1}$

$\frac{d}{dx}(x^4) = 4x^3$

Q.5.2 $u = x^2 + y^2$, *find* (i) $\frac{\partial u}{\partial x}$ (ii) $\frac{\partial u}{\partial y}$

$u = x^2 + y^2$

(i) $\frac{\partial u}{\partial x} = 2x + 0$

$\boxed{\frac{\partial u}{\partial x} = 2x}$

(ii) $\frac{\partial u}{\partial y} = 0 + 2y$

$\boxed{\frac{\partial u}{\partial y} = 2y}$

Q.5.3 $u = x^2 + y^2 + z^2$ „ *find* (i) $\frac{\partial u}{\partial x}$ (ii) $\frac{\partial u}{\partial y}$ (iii) $\frac{\partial u}{\partial z}$

$u = x^2 + y^2 + z^2$

(i) $\frac{\partial u}{\partial x} = 2x + 0 + 0$

$\boxed{\frac{\partial u}{\partial x} = 2x}$

(ii) $\frac{\partial u}{\partial y} = 0 + 2y + 0$

$\boxed{\frac{\partial u}{\partial y} = 2y}$

(iii) $\frac{\partial u}{\partial z} = 0 + 0 + 2z$

$\boxed{\frac{\partial u}{\partial z} = 2z}$

Q.5.4 $V = x^3y^3 + x^2y^2$, *find* (i) $\frac{\partial V}{\partial x}$ (ii) $\frac{\partial V}{\partial y}$ (iii) $\frac{\partial V}{\partial z}$

$$V = x^3y^3 + x^2y^2$$

(i) $\frac{\partial V}{\partial x} = 3x^2y^3 + 2xy^2$

(ii) $\frac{\partial V}{\partial y} = x^3 \times 3y^2 + x^2 2y$

$$\frac{\partial V}{\partial y} = 3x^3y^2 + 2x^2y$$

(iii) $\frac{\partial V}{\partial z} = 0 + 0$

$$\boxed{\frac{\partial V}{\partial z} = 0}$$

Q.5.5 $f(x,y) = x^4 + 6\sqrt{y} - 10$, *find* (i) $\frac{\partial f}{\partial x}$ (ii) $\frac{\partial f}{\partial y}$

$$f(x,y) = x^4 + 6\sqrt{y} - 10$$

(i) $\frac{\partial f}{\partial x} = 4x^3 + 0 - 0$

$$\boxed{\frac{\partial f}{\partial x} = 4x^3}$$

(ii) $\frac{\partial f}{\partial y} = 0 + \frac{6 \times 1}{2\sqrt{y}} - 0$

$$\boxed{\frac{\partial f}{\partial y} = \frac{3}{\sqrt{y}}}$$

$$\boxed{\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}}$$

Q.5.6 $V = \sin(6x - y)$, find (i) $\frac{\partial V}{\partial x}$ (ii) $\frac{\partial V}{\partial y}$

$V = \sin(6x - y)$

(i) $\frac{\partial V}{\partial x} = \cos(6x - y) \frac{\partial}{\partial x}(6x - y)$

$= \cos(6x - y) \times (6 - 0)$

$\therefore \boxed{\frac{\partial V}{\partial x} = 6\cos(6x - y)}$

$\boxed{\frac{d}{dx}(\sin x) = \cos x}$

(ii) $\frac{\partial V}{\partial y} = \cos(6x - y) \frac{\partial}{\partial y}(6x - y)$

$= \cos(6x - y)(0 - 1)$

$\therefore \boxed{\frac{\partial V}{\partial y} = -\cos(6x - y)}$

Q.5.7 $z = \log(x^2 + xy + y^2)$, find (i) $\frac{\partial z}{\partial x}$ (ii) $\frac{\partial z}{\partial y}$

$$z = \log(x^2 + xy + y^2)$$

(i) $$\frac{\partial z}{\partial x} = \frac{1}{(x^2 + xy + y^2)} \times \frac{\partial}{\partial x}(x^2 + xy + y^2)$$

$$\frac{\partial z}{\partial x} = \frac{1}{(x^2 + xy + y^2)} \times (2x + y + 0)$$

$$\boxed{\frac{\partial z}{\partial x} = \frac{2x + y}{x^2 + xy + y^2}}$$

$$\boxed{\frac{d}{dx}(\log x) = \frac{1}{x}}$$

(ii) $$\frac{\partial z}{\partial y} = \frac{1}{(x^2 + xy + y^2)} \times \frac{\partial}{\partial y}(x^2 + xy + y^2)$$

$$= \frac{1}{x^2 + xy + y^2} \times (0 + x + 2y)$$

$$\boxed{\frac{\partial z}{\partial y} = \frac{x + 2y}{x^2 + xy + y^2}}$$

Q.5.8 $z = log\,(tanx + tany)\,, find$ $(i)\frac{\partial z}{\partial x}$ $(ii)\ \frac{\partial z}{\partial y}$

$\frac{d}{dn}(\tan n) = \sec^2 n$

$z = \log(\tan n + \tan y)$

(i) $\frac{\partial z}{\partial n} = \frac{1}{\tan n + \tan y} \times \frac{\partial}{\partial n}(\tan n + \tan y)$

$= \frac{1}{\tan n + \tan y} \times (\sec^2 n + 0)$

$\frac{\partial z}{\partial n} = \frac{\sec^2 n}{\tan n + \tan y}$

Q.5.9 $u = \sin(\sqrt{x} + \sqrt{y})$, find (i) $\frac{\partial u}{\partial x}$ (ii) $\frac{\partial u}{\partial y}$

$$u = \sin(\sqrt{x} + \sqrt{y})$$

$$(i)\ \frac{\partial u}{\partial x} = \cos(\sqrt{x} + \sqrt{y}) \times \frac{\partial}{\partial x}(\sqrt{x} + \sqrt{y})$$

$$= \cos(\sqrt{x} + \sqrt{y}) \times \left(\frac{1}{2\sqrt{x}} + 0\right)$$

$$\boxed{\frac{\partial u}{\partial x} = \frac{1}{2\sqrt{x}} \cos(\sqrt{x} + \sqrt{y})}$$

Q.5.10 $u = y^x$, *find* (i) $\frac{\partial u}{\partial x}$ (ii) $\frac{\partial u}{\partial y}$

$$u = y^x$$

(i) $\frac{\partial u}{\partial x} = y^x \log y$

(ii) $\frac{\partial u}{\partial y} = x y^{x-1}$

$$\frac{d}{dx}(a^x) = a^x \log a$$

$$\frac{d}{dx}(x^n) = n x^{n-1}$$

Q.5.11 $v = e^{x^2+y^2+z^2}$, *find* (i) $\frac{\partial V}{\partial x}$ (ii) $\frac{\partial V}{\partial y}$ (iii) $\frac{\partial V}{\partial z}$

$$V = e^{x^2+y^2+z^2}$$

$$\frac{d(e^x)}{dx} = e^x$$

$$(i)\ \frac{\partial V}{\partial x} = e^{x^2+y^2+z^2} \times \frac{\partial}{\partial x}(x^2+y^2+z^2)$$

$$\frac{\partial V}{\partial x} = e^{x^2+y^2+z^2} \times (2x+0+0)$$

$$\boxed{\frac{\partial V}{\partial x} = 2xe^{x^2+y^2+z^2}}$$

Q.5.12 $u = \tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, find $\frac{\partial u}{\partial x}$

$$u = \tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{1+\left(\frac{x^2+y^2}{x+y}\right)^2} \times \frac{\partial}{\partial x}\left(\frac{x^2+y^2}{x+y}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\frac{1+(x^2+y^2)^2}{(x+y)^2}} \times \left\{\frac{(x+y)\frac{\partial}{\partial x}(x^2+y^2) - (x^2+y^2)\frac{\partial}{\partial x}(x+y)}{(x+y)^2}\right\}$$

$$\frac{\partial u}{\partial x} = \frac{1}{(x+y)^2+(x^2+y^2)^2}\left\{\frac{2x(x+y) - (x^2+y^2)}{(x+y)^2}\right\}$$

$$\frac{\partial u}{\partial x} = \frac{\cancel{(x+y)^2}}{(x+y)^2+(x^2+y^2)^2}\left\{\frac{2x^2+2xy-x^2-y^2}{\cancel{(x+y)^2}}\right\}$$

$$\boxed{\frac{\partial u}{\partial x} = \frac{x^2+2xy-y^2}{(x+y)^2+(x^2+y^2)^2}}$$


Thank
you

AKTU

Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-6

As per New Syllabus 2022-23

Partial Derivative of first order

Part-2

Q.6.1 *If* $z(x+y) = x^2 + y^2$ *, show that* $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.

$$z(x+y) = x^2+y^2$$

$$z = \frac{x^2+y^2}{x+y}$$

$$\frac{\partial z}{\partial x} = \frac{(x+y)\frac{\partial}{\partial x}(x^2+y^2) - (x^2+y^2)\frac{\partial}{\partial x}(x+y)}{(x+y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{(x+y)(2x+0) - (x^2+y^2)(1+0)}{(x+y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{2x^2+2xy-x^2-y^2}{(x+y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{x^2+2xy-y^2}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(x+y)\frac{\partial}{\partial y}(x^2+y^2) - (x^2+y^2)\frac{\partial}{\partial y}(x+y)}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(x+y)(0+2y) - (x^2+y^2)(0+1)}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{2xy+2y^2-x^2-y^2}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{y^2+2xy-x^2}{(x+y)^2}$$

$$\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2 = 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)$$

LHS

$$\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2$$

$$=\left[\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}\right]^2$$

$$=\left[\frac{x^2+2xy-y^2-y^2-2xy+x^2}{(x+y)^2}\right]^2$$

$$=\left[\frac{2x^2-2y^2}{(x+y)^2}\right]^2=\left[\frac{2(x^2-y^2)}{(x+y)^2}\right]^2$$

$$=\left[\frac{2(x+y)(x-y)}{(x+y)^2}\right]^2$$

$$=\frac{4(x-y)^2}{(x+y)^2}$$

RHS

$$=4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)$$

$$=4\left[1-\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}\right]$$

$$=4\left[\frac{(x+y)^2-x^2-2xy+y^2-y^2-2xy+x^2}{(x+y)^2}\right]$$

$$=4\left[\frac{x^2+y^2+2xy-2xy-2xy}{(x+y)^2}\right]$$

$$=4\left[\frac{x^2+y^2-2xy}{(x+y)^2}\right]$$

$$=\frac{4(x-y)^2}{(x+y)^2}$$

LHS = RHS

Hence proved

Q.6.2 *If* $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$ *then find the value of* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$.

$$u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-\left(\frac{x}{y}\right)^2}} \times \frac{\partial}{\partial x}\left(\frac{x}{y}\right) + \frac{1}{1+\left(\frac{y}{x}\right)^2} \times \frac{\partial}{\partial x}\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \times \frac{1}{y} + \frac{1}{1+\frac{y^2}{x^2}} \times \left(\frac{-y}{x^2}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{\frac{y^2-x^2}{y^2}}} \times \frac{1}{y} + \frac{1}{\frac{x^2+y^2}{x^2}} \times \left(\frac{-y}{x^2}\right)$$

$$\frac{\partial u}{\partial x} = \frac{y}{\sqrt{y^2-x^2}} \times \frac{1}{y} - \frac{x^2 y}{(x^2+y^2)x^2}$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{1-\left(\frac{x}{y}\right)^2}} \times \frac{\partial}{\partial y}\left(\frac{x}{y}\right) + \frac{1}{1+\left(\frac{y}{x}\right)^2} \times \frac{\partial}{\partial y}\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{\frac{y^2-x^2}{y^2}}} \times \left(\frac{-x}{y^2}\right) + \frac{1}{\frac{x^2+y^2}{x^2}} \times \frac{1}{x}$$

$$\frac{\partial u}{\partial y} = \frac{y}{\sqrt{y^2-x^2}} \times \left(\frac{-x}{y^2}\right) + \frac{x^2}{x^2+y^2} \times \frac{1}{x}$$

$$\frac{\partial u}{\partial y} = \frac{-x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = x\left[\frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2}\right] + y\left[\frac{-x}{y\sqrt{y^2-x^2}} + \frac{x}{x^2+y^2}\right]$$

$$= \frac{x}{\sqrt{y^2-x^2}} - \frac{xy}{x^2+y^2} - \frac{xy}{y\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2}$$

$$= \frac{x}{\sqrt{y^2-x^2}} - \frac{xy}{x^2+y^2} - \frac{x}{\sqrt{y^2-x^2}} + \frac{xy}{x^2+y^2}$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0}$$

Q.6.3 *If $u(x, y, z) = \log(\tan x + \tan y + \tan z)$, show that*

$$\text{Sin } 2x\frac{\partial u}{\partial x} + \text{Sin } 2y\frac{\partial u}{\partial y} + \text{Sin } 2z\frac{\partial u}{\partial z} = 2$$

$u(x, y, z) = \log(\tan x + \tan y + \tan z)$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \times \frac{\partial}{\partial x}(\tan x + \tan y + \tan z)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z}(\sec^2 x + 0 + 0)$$

$$\boxed{\frac{\partial u}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y + \tan z}}$$

$$\frac{\partial u}{\partial y} = \frac{1}{(\tan x + \tan y + \tan z)}\frac{\partial}{\partial y}(\tan x + \tan y + \tan z)$$

$$\boxed{\frac{\partial u}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z}}$$

$$\frac{\partial u}{\partial z} = \frac{1}{\tan x + \tan y + \tan z} \times \frac{\partial}{\partial z}(\tan x + \tan y + \tan z)$$

$$\boxed{\frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}}$$

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$$

$$= \frac{2\tan x + 2\tan y + 2\tan z}{\tan x + \tan y + \tan z}$$

LHS

$$= \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z}$$

$$= \frac{\sin 2x \sec^2 x}{\tan x + \tan y + \tan z} + \frac{\sin 2y \sec^2 y}{\tan x + \tan y + \tan z} + \frac{\sin 2z \sec^2 z}{\tan x + \tan y + \tan z}$$

$$= \frac{2 \sin x \cos x \times \frac{1}{\cos^2 x}}{\tan x + \tan y + \tan z} + \frac{2 \sin y \cos y \times \frac{1}{\cos^2 y}}{\tan x + \tan y + \tan z} + \frac{2 \sin z \cos z \frac{1}{\cos^2 z}}{\tan x + \tan y + \tan z}$$

$$= \frac{2\tan x}{\tan x + \tan y + \tan z} + \frac{2\tan y}{\tan x + \tan y + \tan z} + \frac{2\tan z}{\tan x + \tan y + \tan z}$$

$$= \frac{2(\tan x + \tan y + \tan z)}{(\tan x + \tan y + \tan z)}$$

$$= 2$$

LHS = RHS

Hence proved


Thank
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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-7

As per New Syllabus 2022-23



Partial Derivative of higher order

Part-1

Partial Derivative of higher order

If $z = f(x,y)$ then $\left(\frac{\partial z}{\partial x} \text{ or } \frac{\partial f}{\partial x} \text{ or } f_x \text{ or } u_x\right)$ and $\left(\frac{\partial z}{\partial y} \text{ or } \frac{\partial f}{\partial y} \text{ or } f_y \text{ or } u_y\right)$ being the function of x and y, can further be diffrentiated partially with respect to x and y.

① $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2}$ OR $\frac{\partial^2 f}{\partial x^2}$ OR f_{xx} OR u_{xx}

② $\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2}$ OR $\frac{\partial^2 f}{\partial y^2}$ OR f_{yy} OR u_{yy}

③ $\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x}$ OR $\frac{\partial^2 f}{\partial y \partial x}$ OR f_{yx} OR u_{yx}

④ $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y}$ OR $\frac{\partial^2 f}{\partial x \partial y}$ OR f_{xy} OR u_{xy}

Note

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$$

Q.7.1

If $z = \log(e^x + e^y)$, show that $rt - s^2 = 0$, where $r = \dfrac{\partial^2 z}{\partial x^2}, t = \dfrac{\partial^2 z}{\partial y^2}, s = \dfrac{\partial^2 z}{\partial x \partial y}$.

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$z = \log(e^x + e^y)$

Partially Diff wrt x

$$\frac{\partial z}{\partial x} = \frac{1}{e^x + e^y} \times (e^x + 0)$$

$$\frac{\partial z}{\partial x} = \frac{e^x}{e^x + e^y}$$

Again partially diff wrt x

$$\frac{\partial^2 z}{\partial x^2} = \frac{(e^x + e^y)\frac{\partial}{\partial x}(e^x) - e^x \frac{\partial}{\partial x}(e^x + e^y)}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{(e^x + e^y)e^x - e^x \times (e^x + 0)}{(e^x + e^y)^2}$$

$$r = \frac{\partial^2 z}{\partial x^2} = \frac{e^x \times e^x + e^x \times e^y - e^x \times e^x}{(e^x + e^y)^2}$$

$$r = \frac{e^{2x} + e^{x+y} - e^{2x}}{(e^x + e^y)^2}$$

$$\boxed{r = \frac{e^{x+y}}{(e^x + e^y)^2}} \quad \text{—①}$$

$z = \log(e^x + e^y)$

$$\frac{\partial z}{\partial y} = \frac{1}{e^x + e^y} \times (0 + e^y)$$

$$\frac{\partial z}{\partial y} = \frac{e^y}{e^x + e^y}$$

Again partially diff wrt y

$$\frac{\partial^2 z}{\partial y^2} = \frac{(e^x + e^y)\frac{\partial}{\partial y}(e^y) - e^y \frac{\partial}{\partial y}(e^x + e^y)}{(e^x + e^y)^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{(e^x+e^y)e^y - e^y(0+e^y)}{(e^x+e^y)^2}$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{e^x e^y + \cancel{e^y \times e^y} - \cancel{e^y \times e^y}}{(e^x+e^y)^2}$$

$$\boxed{t = \frac{e^{x+y}}{(e^x+e^y)^2}} \quad \text{—(2)}$$

$$z = \log(e^x+e^y)$$

Partially diff. wrt y

$$\frac{\partial z}{\partial y} = \frac{1}{e^x+e^y} \times (0+e^y)$$

$$\frac{\partial z}{\partial y} = \frac{e^y}{e^x+e^y}$$

Again partially diff wrt x

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{(e^x+e^y)\frac{\partial}{\partial x}(e^y) - e^y \frac{\partial}{\partial x}(e^x+e^y)}{(e^x+e^y)^2}$$

$$s = \frac{0 - e^y \times e^x}{(e^x+e^y)^2}$$

$$\boxed{s = \frac{-e^{x+y}}{(e^x+e^y)^2}} \quad \text{—(3)}$$

$$rt - s^2 = 0$$

LHS

$$= rt - s^2$$

$$= \frac{e^{x+y}}{(e^x+e^y)^2} \times \frac{e^{x+y}}{(e^x+e^y)^2} - \left(\frac{-e^{x+y}}{(e^x+e^y)^2}\right)^2$$

$$= \left[\frac{e^{x+y}}{(e^x+e^y)^2}\right]^2 - \left[\frac{e^{x+y}}{(e^x+e^y)^2}\right]^2$$

$$= 0$$

$$\boxed{\text{LHS} = \text{RHS}}$$

Hence proved

Q.7.2 If $u = (x^2 + y^2 + z^2)^{-1/2}$, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.

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$u = (x^2+y^2+z^2)^{-\frac{1}{2}}$

$\frac{\partial u}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}} \times \frac{\partial}{\partial x}(x^2+y^2+z^2)$

$\frac{\partial u}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}} \times 2x$

$\frac{\partial u}{\partial x} = -x(x^2+y^2+z^2)^{-\frac{3}{2}}$

$\frac{\partial^2 u}{\partial x^2} = -\left[x \times -\frac{3}{2}(x^2+y^2+z^2)^{-\frac{5}{2}} \times 2x + (x^2+y^2+z^2)^{-\frac{3}{2}}\right]$

$\frac{\partial^2 y}{\partial x^2} = 3x^2(x^2+y^2+z^2)^{-\frac{5}{2}} - (x^2+y^2+z^2)^{-\frac{3}{2}}$ — (1)

$u = (x^2+y^2+z^2)^{-\frac{1}{2}}$

$\frac{\partial u}{\partial y} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}} \times \frac{\partial}{\partial y}(x^2+y^2+z^2)$

$\frac{\partial u}{\partial y} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}} \times 2y$

$\frac{\partial u}{\partial y} = -y(x^2+y^2+z^2)^{-\frac{3}{2}}$

$\frac{\partial^2 u}{\partial y^2} = -\left[y\left(-\frac{3}{2}\right)(x^2+y^2+z^2)^{-\frac{5}{2}} \times 2y + (x^2+y^2+z^2)^{-\frac{3}{2}}\right]$

$\frac{\partial^2 y}{\partial y^2} = 3y^2(x^2+y^2+z^2)^{-\frac{5}{2}} - (x^2+y^2+z^2)^{-\frac{3}{2}}$ — (2)

$$u = (x^2+y^2+z^2)^{-\frac{1}{2}}$$

$$\frac{\partial^2 u}{\partial z^2} = 3z^2(x^2+y^2+z^2)^{-\frac{5}{2}} - (x^2+y^2+z^2)^{-\frac{3}{2}} \quad -③$$

Adding eqn ①, ② and ③

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2} = \underbrace{3x^2(x^2+y^2+z^2)^{-\frac{5}{2}} - (x^2+y^2+z^2)^{-\frac{3}{2}}}+\underbrace{3y^2(x^2+y^2+z^2)^{-\frac{5}{2}} - (x^2+y^2+z^2)^{-\frac{3}{2}}}+3z^2(x^2+y^2+z^2)^{-\frac{5}{2}}$$

$$\underbrace{-(x^2+y^2+z^2)^{-\frac{3}{2}}}$$

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2} = 3(x^2+y^2+z^2)^{-\frac{5}{2}}(x^2+y^2+z^2)^{1} - 3(x^2+y^2+z^2)^{-\frac{3}{2}}$$

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2} = \cancel{3(x^2+y^2+z^2)^{-\frac{3}{2}}} - \cancel{3(x^2+y^2+z^2)^{-\frac{3}{2}}}$$

$$\boxed{\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2} = 0}$$

Hence proved

Q.7.3

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Prove that

If $u = x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$, ~~find the value of~~ $\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$

$$u = x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$$

Partially diff. w.r.t. y

$$\frac{\partial u}{\partial y} = x^2 \times \frac{1}{1+\left(\frac{y}{x}\right)^2} \times \frac{1}{x} - \left[y^2 \times \frac{1}{1+\left(\frac{x}{y}\right)^2} \times \left(-\frac{x}{y^2}\right) + \left(\tan^{-1}\frac{x}{y}\right) \times 2y\right]$$

$$\frac{\partial u}{\partial y} = \frac{x \times x^2}{x^2+y^2} - \left[\frac{-y^2 x}{x^2+y^2} + 2y \tan^{-1}\left(\frac{x}{y}\right)\right]$$

$$\frac{\partial u}{\partial y} = \frac{x^3}{x^2+y^2} + \frac{xy^2}{x^2+y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial y} = \frac{x^3 + xy^2}{x^2+y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial y} = \frac{x(x^2+y^2)}{(x^2+y^2)} - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial y} = x - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

Again partially diff. w.r.t. x

$$\frac{\partial^2 u}{\partial x \partial y} = 1 - 2y \times \frac{1}{1+\frac{x^2}{y^2}} \times \frac{1}{y}$$

$$\frac{\partial^2 u}{\partial x \partial y} = 1 - \frac{2}{1 + \frac{x^2}{y^2}}$$

$$\frac{\partial^2 u}{\partial x \partial y} = 1 - \frac{2y^2}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 + y^2 - 2y^2}{x^2 + y^2}$$

$$\boxed{\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}}$$

Hence proved

Practice Q.7.1

If $u = \log(x^2 + y^2 + z^2)$, show that $x\dfrac{\partial^2 u}{\partial y \partial z} = y\dfrac{\partial^2 u}{\partial z \partial x} = z\dfrac{\partial^2 u}{\partial x \partial y}$.

Practice Q.7.2

If $u = \log\sqrt{x^2 + y^2 + z^2}$, show that $(x^2 + y^2 + z^2)\left(\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2}\right) = 1$.

Practice Q.7.3

If $u = e^{xyz}$, prove that $\dfrac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)u$.

Practice Q.7.4

If $x = r\cos\theta, y = r\sin\theta$, prove that

(i) $\dfrac{\partial^2 r}{\partial x^2} \cdot \dfrac{\partial^2 r}{\partial y^2} = \left(\dfrac{\partial^2 r}{\partial x \partial y}\right)^2$

(ii) $\dfrac{\partial^2 r}{\partial x^2} + \dfrac{\partial^2 r}{\partial y^2} = \dfrac{1}{r}\left[\left(\dfrac{\partial r}{\partial x}\right)^2 + \left(\dfrac{\partial r}{\partial y}\right)^2\right]$.


Thank
you

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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-8

As per New Syllabus 2022-23



Partial Derivative of higher order

Part-2

Q.8.1 If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

$u = \log(x^3+y^3+z^3-3xyz)$

Partially Diff. wrt x

$$\frac{\partial u}{\partial x} = \frac{1 \times (3x^2+0+0-3yz)}{(x^3+y^3+z^3-3xyz)}$$

$$\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz} \quad -(1)$$

Partially diff u wrt y

$$\frac{\partial u}{\partial y} = \frac{1 \times (0+3y^2+0-3xz}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz} \quad -(2)$$

Partially diff. u wrt z

$$\frac{\partial u}{\partial z} = \frac{1 \times (0+0+3z^2-3xy)}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz} \quad -(3)$$

Adding eqn (1), (2) and (3)

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz} + \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz}$$

$$+ \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3x^2-3yz+3y^2-3xz+3z^2-3xy}{x^3+y^3+z^3-3xyz}=\frac{3(x^2+y^2+z^2-yz-xz-xy)}{x^3+y^3+z^3-3xyz}$$

$$\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)}$$

$$\boxed{\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3}{(x+y+z)}}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=-\frac{9}{(x+y+z)^2}$$

LHS

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)u$$

$$=\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$$

$$=\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\frac{3}{(x+y+z)}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\frac{3}{x+y+z}$$

$$= \frac{\partial}{\partial x}\left(\frac{3}{x+y+z}\right)+\frac{\partial}{\partial y}\left(\frac{3}{x+y+z}\right)+\frac{\partial}{\partial z}\left(\frac{3}{x+y+z}\right)$$

$$= \frac{-3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2}$$

$$= \frac{-3-3-3}{(x+y+z)^2}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$\frac{d}{dx}\left(\frac{1}{x+z+3}\right) = -\frac{1}{(x+z+3)^2}$$

Practice Q.

If $u = f(r)$, where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$


Thank
you

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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-9

As per New Syllabus 2022-23

Euler's Theorem

Part-1

Homogeneous Functions

A function f(x,y) is said to be homogeneous of degree n in the variables x and y if it can be expressed in the form $x^n \phi\left(\frac{y}{x}\right)$ or $y^n \phi\left(\frac{x}{y}\right)$

Alternative test

(i) A function f(x,y) will be homogeneous of degree n , if

$$f(tx, ty) = t^n f(x, y)$$

(ii) A function f(x,y,z) will be homogeneous of degree n , if

$$f(tx, ty, tz) = t^n f(x, y, z)$$

Euler's Theorem

State and prove Euler's theorem for partial differentiation of a homogeneous function f (x, y).

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Statement : If u is a homogeneous function of degree n in x and y, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

Proof : Since u is a homogeneous function of degree n in x and y, it can be expressed as

$$u = x^n f\left(\frac{y}{x}\right)$$

$$u = x^n f\left(\frac{y}{x}\right)$$

Partially diff. w.r.t x

$$\frac{\partial u}{\partial x} = x^n f'\left(\frac{y}{x}\right) \times \left(\frac{-y}{x^2}\right) + f\left(\frac{y}{x}\right) \times n \times x^{n-1}$$

$$\frac{\partial u}{\partial x} = \frac{-y}{x^2} \times x^n f'\left(\frac{y}{x}\right) + n x^{n-1} f\left(\frac{y}{x}\right)$$

Multiply both side by x

$$x\frac{\partial u}{\partial x} = \frac{-xy}{x^2} \times x^n f'\left(\frac{y}{x}\right) + n x \times x^{n-1} f\left(\frac{y}{x}\right)$$

$$x\frac{\partial u}{\partial x} = -x^{n-1} y f'\left(\frac{y}{x}\right) + n x^n f\left(\frac{y}{x}\right) \quad -①$$

Again diff. partially w.r.t y

$$\frac{\partial u}{\partial y} = x^n f'\left(\frac{y}{x}\right) \times \frac{1}{x}$$

Multiply both side by y

$$y\frac{\partial u}{\partial y} = x^n y f'\left(\frac{y}{x}\right) \times \frac{1}{x}$$

$$y\frac{\partial u}{\partial y} = x^{n-1} y f'\left(\frac{y}{x}\right) \quad -②$$

Adding equation ① and ②

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -x^{n-1} y f'\left(\frac{y}{x}\right) + n x^n f\left(\frac{y}{x}\right) + x^{n-1} y f'\left(\frac{y}{x}\right)$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\left(x^n f\left(\frac{y}{x}\right)\right)$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu}$$

Hence proved

Note-1 : If u is a homogeneous function of degree n in x , y and z, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$$

Hence Euler's theorem can be extended to any number of variables

Note-2 : If u is a homogeneous function of degree n in x and y then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

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Q.9.1 Verify Euler's theorem for the following function

$$z = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$$

$$u(x,y) = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$$

$$u(tx,ty) = \frac{(tx)^{\frac{1}{3}} + (ty)^{\frac{1}{3}}}{(tx)^{\frac{1}{2}} + (ty)^{\frac{1}{2}}}$$

$$u(tx,ty) = \frac{t^{1/3}x^{1/3} + t^{1/3}y^{1/3}}{t^{1/2}x^{1/2} + t^{1/2}y^{1/2}}$$

$$u(tx,ty) = \frac{t^{1/3}\left(x^{1/3} + y^{1/3}\right)}{t^{1/2}\left(x^{1/2} + y^{1/2}\right)}$$

$$u(tx,ty) = t^{-1/6}\left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)$$

$$u(tx,ty) = t^{-1/6}\,u(x,y)$$

$\therefore$ u is a homogeneous function in which $n = -\frac{1}{6}$

By Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{6}u} \quad \text{(A)}$$

$$u(x,y) = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$$

Partially diff w.r.t. x

$$\frac{\partial u}{\partial x} = \frac{\left(x^{1/2} + y^{1/2}\right)\left(\frac{1}{3}x^{-2/3}\right) - \left(x^{1/3} + y^{1/3}\right)\left(\frac{1}{2}x^{-1/2}\right)}{\left(x^{1/2} + y^{1/2}\right)^2}$$

$$\frac{\partial u}{\partial x} = \frac{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}}\right)\left(\frac{1}{3}x^{-2/3}\right)}{\left(x^{1/2} + y^{1/2}\right)^2} - \frac{\left(x^{\frac{1}{3}} + y^{1/3}\right)\left(\frac{1}{2}x^{-1/2}\right)}{\left(x^{1/2} + y^{1/2}\right)^2}$$

$$\frac{\partial u}{\partial x} = \frac{x^{-2/3}}{3\left(x^{1/2} + y^{1/2}\right)} - \frac{\left(x^{1/3} + y^{1/3}\right)\left(x^{-1/2}\right)}{2\left(x^{1/2} + y^{1/2}\right)\left(x^{1/2} + y^{1/2}\right)}$$

$$\frac{\partial u}{\partial x} = \frac{x^{-2/3}}{3(x^{1/2}+y^{1/2})} - \frac{u\,x^{-1/2}}{2(x^{1/2}+y^{1/2})}$$

Multiply both sides by x

$$x\frac{\partial u}{\partial x} = \boxed{\frac{x^{1/3}}{3(x^{1/2}+y^{1/2})}} - \frac{u \times x^{1/2}}{2(x^{1/2}+y^{1/2})} \quad -(1)$$

Partially diff u wrt y

$$u = \frac{x^{1/3}+y^{1/3}}{x^{1/2}+y^{1/2}}$$

$$\frac{\partial u}{\partial y} = \frac{(x^{\frac{1}{2}}+y^{\frac{1}{2}})(\frac{1}{3}y^{-\frac{2}{3}}) - (\frac{1}{x^3}+y^{\frac{1}{3}})(\frac{1}{2}y^{-1/2})}{(x^{1/2}+y^{1/2})^2}$$

$$\frac{\partial u}{\partial y} = \frac{(x^{\frac{1}{2}}+y^{\frac{1}{2}})(y^{-2/3})}{3(x^{1/2}+y^{1/2})^2} - \frac{(x^{1/3}+y^{1/3})(y^{-1/2})}{2(x^{1/2}+y^{1/2})^2}$$

$$\frac{\partial u}{\partial y} = \frac{y^{-2/3}}{3(x^{1/2}+y^{1/2})} - \boxed{\frac{(x^{1/3}+y^{1/3})(y^{1/2})}{(x^{1/2}+y^{1/2})\,2(x^{1/2}+y^{1/2})}}$$

$$\frac{\partial u}{\partial y} = \frac{y^{-2/3}}{3(x^{1/2}+y^{1/2})} - \frac{u\,y^{-1/2}}{2(x^{1/2}+y^{1/2})}$$

Multiply both sides by y

$$y\frac{\partial u}{\partial y} = \boxed{\frac{y^{1/3}}{3(x^{1/2}+y^{1/2})}} - \frac{u \times y^{1/2}}{2(x^{1/2}+y^{1/2})} \quad -(2)$$

Adding (1) and (2)

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{x^{1/3}}{3(x^{1/2}+y^{1/2})} + \frac{y^{1/3}}{3(x^{1/2}+y^{1/2})} -$$

$$\frac{u \times x^{1/2}}{2(x^{1/2}+y^{1/2})} - \frac{u \times y^{1/2}}{2(x^{1/2}+y^{1/2})}$$

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{x^{1/3}+y^{1/3}}{3(x^{1/2}+y^{1/2})}-\frac{u(x^{1/2}+y^{1/2})}{2(x^{1/2}+y^{1/2})}$$

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{u}{3}-\frac{u}{2}$$

$$=\frac{2u-3u}{6}$$

$$\boxed{x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=-\frac{u}{6}} \quad \text{— (B)}$$

From (A) and (B)

Euler's theorem is verified

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Q.9.2 *If* $u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$, *find the value of* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$

$$u(x,y) = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$$

$$u(tx,ty) = \sin^{-1}\left(\frac{tx}{ty}\right) + \tan^{-1}\left(\frac{ty}{tx}\right)$$

$$u(tx,ty) = t^0 \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$$

$$u(tx,ty) = t^0 u(x,y)$$

$\therefore$ u is a Homogeneous function inwhich $n = 0$ (degree)

By Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu = 0 \times u$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0}$$

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Q.9.3 If $u = \cos\left(\frac{xy + yz + zx}{x^2 + y^2 + z^2}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$

$$u(x, y, z) = \cos\left(\frac{xy + yz + zx}{x^2 + y^2 + z^2}\right)$$

$$u(tx, ty, tz) = \cos\left(\frac{tx \times ty + ty \times tz + tz \times tx}{(tx)^2 + (ty)^2 + (tz)^2}\right)$$

$$= \cos\left[\frac{t^2xy + t^2yz + t^2zx}{t^2x^2 + t^2y^2 + t^2z^2}\right]$$

$$= \cos\left[\frac{t^2(xy + yz + zx)}{t^2(x^2 + y^2 + z^2)}\right]$$

$$u(tx, ty, tz) = t^0 \cos\left[\frac{xy + yz + zx}{x^2 + y^2 + z^2}\right]$$

$$\boxed{u(tx, ty, tz) = t^0 u(x, y, z)}$$

Hence

u is a Homogeneous function in which $n = 0$ (degree)

By Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu = 0 \times u$$

$$= 0$$

Hence proved

Q.9.4 If $u(x, y) = (\sqrt{x} + \sqrt{y})^5$, find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$.

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$$u(x,y) = (\sqrt{x} + \sqrt{y})^5$$

$$u(tx, ty) = (\sqrt{tx} + \sqrt{ty})^5$$

$$= \left[\left(\sqrt{t} \times \sqrt{x} + \sqrt{t}\sqrt{y}\right)\right]^5$$

$$= \left[\sqrt{t}\left(\sqrt{x} + \sqrt{y}\right)\right]^5$$

$$u(tx, ty) = (\sqrt{t})^5 \times (\sqrt{x} + \sqrt{y})^5$$

$$\boxed{u(tx, ty) = t^{5/2} u(x, y)}$$

Hence u is a Homogeneous function in which

$$n = \frac{5}{2} \text{ (Degree)}$$

By Euler's theorem

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$= \frac{5}{2}\left(\frac{5}{2} - 1\right)u$$

$$= \frac{5}{2} \times \frac{3}{2} \times u$$

$$= \frac{15}{4} u$$


Thank
you

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Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-10

As per New Syllabus 2022-23

Euler's Theorem

Part-2

[Deduction From Euler's Theorem]

Where We apply Deductions from Euler's Theorem

A. Inverse Trigonometric function (Initially not Homogeneous)

1. $u = sin^{-1}(\quad)$
2. $u = cos^{-1}(\quad)$
3. $u = tan^{-1}(\quad)$
4. $u = cot^{-1}(\quad)$
5. $u = cosec^{-1}(\quad)$
6. $u = sec^{-1}(\quad)$

B. Logarithmic function (Initially not Homogeneous)

$u = log(\quad)$

Euler's theorem

1. If u is a homogeneous function of degree n in x and y, then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

2. If u is a homogeneous function of degree n in x , y and z then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$$

3. If u is a homogeneous function of degree n in x and y, then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

Deduction from Euler's theorem

1. If z is a homogeneous function of x and y, of degree n and z = f(u), then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{n\,F(u)}{F'(u)}$$

2. If z is a homogeneous function of x , y and z of degree n and z = f(u), then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{n\,F(u)}{F'(u)}$$

3. If z is a homogeneous function of x and y, of degree n and z = f(u), then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = \phi(u)\left[\phi'(u) - 1\right]$$

Q.10.1 If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$.

$u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$

u is not a Homogeneous function

Let $\sin u = \frac{x^2+y^2}{x+y} = f(x,y)$

Where $F(u) = \sin u$

$F'(u) = \cos u$

$f(x,y) = \frac{x^2+y^2}{x+y}$

$f(tx,ty) = \frac{(tx)^2+(ty)^2}{tx+ty}$

$= \frac{t^2x^2+t^2y^2}{tx+ty}$

$= \frac{t^2(x^2+y^2)}{t(x+y)}$

$f(tx,ty) = t\left(\frac{x^2+y^2}{x+y}\right)$

$\boxed{f(tx,ty) = t\,f(x,y)}$

$\sin u$ is a homogeneous function of degree (1)

By Deduction from Euler's theorem

$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{n\,F(u)}{F'(u)}$

$= 1 \times \frac{\sin u}{\cos u}$

$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u}$

Hence proved

Q.10.2 If $u = \cos^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + \dfrac{1}{2}\cot u = 0$.

$$u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$$

u is not a Homogeneous function

Let $\cos u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}} = f(x,y)$

Where $F(u) = \cos u$

$F'(u) = -\sin u$

$$f(x,y) = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$f(tx,ty) = \frac{tx+ty}{\sqrt{tx}+\sqrt{ty}}$$

$$= \frac{t(x+y)}{\sqrt{t}\times\sqrt{x}+\sqrt{t}\times\sqrt{y}}$$

$$= \frac{t(x+y)}{\sqrt{t}(\sqrt{x}+\sqrt{y})}$$

$$= t^{1/2}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$$

$$\boxed{f(tx,ty) = t^{1/2} f(x,y)}$$

cosu is a Homogeneous function of degree $1/2$

By Deduction from Euler's Theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{F(u)}{F'(u)}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\times\frac{\cos u}{-\sin u}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{2}\cot u$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + \frac{1}{2}\cot u = 0}$$

Q.10.3 If $u = \log_e\left(\frac{x^4+y^4}{x+y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$

$u = \log_e\left(\frac{x^4+y^4}{x+y}\right)$

u is not a Homogeneous function

Let $e^u = \frac{x^4+y^4}{x+y} = f(x,y)$

where $F(u) = e^u$

$F'(u) = e^u$

$f(x,y) = \frac{x^4+y^4}{x+y}$

$f(tx,ty) = \frac{(tx)^4+(ty)^4}{tx+ty}$

$f(tx,ty) = \frac{t^4x^4+t^4y^4}{tx+ty}$

$= \frac{t^{4-1}}{t}\left(\frac{x^4+y^4}{x+y}\right)$

$f(tx,ty) = t^3 f(x,y)$

e^u is a Homogeneous function of degree (3)

By Deduction from Euler's theorem

$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{n\,F(u)}{F'(u)}$

$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3 \times \frac{e^u}{e^u}$

$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3}$

Hence proved

Q.10.4 Show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=2\tan u$, where $u=\sin^{-1}\left(\frac{x^3+y^3+z^3}{ax+by+cz}\right)$.

$u=\sin^{-1}\left(\frac{x^3+y^3+z^3}{ax+by+cz}\right)$

u is not a Homogeneous function

Let $\sin u=\frac{x^3+y^3+z^3}{ax+by+cz}=f(x,y,z)$

When $F(u)=\sin u$

$F'(u)=\cos u$

$f(x,y,z)=\frac{x^3+y^3+z^3}{ax+by+cz}$

$f(tx,ty,tz)=\frac{(tx)^3+(ty)^3+(tz)^3}{axtx+bxty+cxtz}$

$=\frac{t^3}{t}\left(\frac{x^3+y^3+z^3}{ax+by+cz}\right)$

$f(tx,ty,tz)=t^2f(x,y,z)$

$\sin u$ is a Homogeneous function of degree (2)

By Deduction from Euler's theorem

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=\frac{n\,F(u)}{F'(u)}$

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=2\frac{\sin u}{\cos u}$

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=2\tan u$

Hence proved

Q.10.5 If $u = \sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + 3\tan u = 0$

$$u = \sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$$

u is not a Homogeneous function

Let $\sin u = \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} = f(x,y,z)$

Where $F(u) = \sin u$

$F'(u) = \cos u$

$$f(x,y,z) = \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$$

$$f(tx,ty,tz) = \frac{tx+2ty+3tz}{\sqrt{(tx)^8+(ty)^8+(tz)^8}}$$

$$= \frac{t(x+2y+3z)}{\sqrt{t^8(x^8+y^8+z^8)}}$$

$$= \frac{t}{t^4}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$$

$$f(tx,ty,tz) = t^{-3} f(x,y,z)$$

$\sin u$ is a Homogeneous function of degree (-3)

By Deduction from Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{n\,F(u)}{F'(u)}$$

$$= -3\,\frac{\sin u}{\cos u}$$

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = -3\tan u$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + 3\tan u = 0}$$

Hence proved

Q.10.6 If $u = \log\left(\frac{x^5 + y^5 + z^5}{x^2 + y^2 + z^2}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 3$.

$$u = \log\left(\frac{x^5 + y^5 + z^5}{x^2 + y^2 + z^2}\right)$$

u is not a Homogeneous function

Let $e^u = \frac{x^5 + y^5 + z^5}{x^2 + y^2 + z^2} = f(x, y, z)$

Where $F(u) = e^u$

$F'(u) = e^u$

Q.10.7 If $u = \sin^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that

(i) $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{2}\tan u$

(ii) $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = -\dfrac{\sin u \cos 2u}{4\cos^3 u}$.

$u = \sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$

u is not a Homogeneous function

Let $\sin u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}} = f(x,y)$

where $F(u) = \sin u$

$F'(u) = \cos u$

$f(x,y) = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$

$f(tx,ty) = \dfrac{tx+ty}{\sqrt{tx}+\sqrt{ty}}$

$f(tx,ty) = \dfrac{t}{\sqrt{t}}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$

$f(tx,ty) = t^{1/2} f(x,y)$

$\sin u$ is a Homogeneous of degree $\left(\frac{1}{2}\right)$

By Deduction from Eulers theorem

$x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{n\,F(u)}{F'(u)}$

$= \frac{1}{2} \times \dfrac{\sin u}{\cos u}$

$x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \frac{1}{2}\tan u$

Where $\phi(u) = \frac{1}{2}\tan u$

$$\phi'(u) = \frac{1}{2}\sec^2 u$$

By Deduction from Euler's theorem

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \phi(u)\left[\phi'(u) - 1\right]$$

$$= \frac{1}{2}\tan u\left[\frac{1}{2}\sec^2 u - 1\right]$$

$$= \frac{1}{2} \times \frac{\sin u}{\cos u}\left[\frac{1}{2\cos^2 u} - 1\right]$$

$$= \frac{1}{2} \times \frac{\sin u}{\cos u}\left(\frac{1 - 2\cos^2 u}{2\cos^2 u}\right)$$

$$= \frac{1}{2} \times \frac{\sin u}{\cos u} \times \frac{(-\cos 2u)}{2\cos^2 u}$$

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} =$$

$$= -\frac{1}{4}\frac{\sin u \cos 2u}{\cos^3 u}$$

Hence proved

Q.10.8 If $u = \sec^{-1}\left(\frac{x^3 - y^3}{x + y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\cot u$. Also evaluate $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

Q.10.9 If $u = \operatorname{cosec}^{-1}\left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)^{1/2}$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}(13 + \tan^2 u)$.

$$u = \operatorname{cosec}^{-1}\left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)^{1/2}$$

u is not a Homogeneous function

Let $\operatorname{cosec} u = \left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)^{1/2} = f(x,y)$

Where $F(u) = \operatorname{cosec} u$

$$F'(u) = -\operatorname{cosec} u \cot u$$

$$f(x,y) = \left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)^{1/2}$$

$$f(tx,ty) = \left[\frac{(tx)^{1/2}+(ty)^{1/2}}{(tx)^{1/3}+(ty)^{1/3}}\right]^{1/2}$$

$$= \left[\frac{t^{1/2}(x^{1/2}+y^{1/2})}{t^{1/3}(x^{1/3}+y^{1/3})}\right]^{1/2}$$

$$= \left[t^{1/6}\left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)\right]^{\frac{1}{2}}$$

$$= t^{1/12}\left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)^{1/2} = t^{1/12} f(x,y)$$

Q.10.10 If $z = x^4 y^2 \sin^{-1}\left(\frac{x}{y}\right) + \log x - \log y$, show that $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 6x^4y^2 \sin^{-1}\left(\frac{x}{y}\right)$.

Practice Q.10.1 If $u = (x^{1/4} + y^{1/4})(x^{1/5} + y^{1/5})$, apply Euler's theorem to find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$. [M.T.U. (SUM) 2011]

Practice Q.10.2 $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u = 2 \cos 3u \sin u.$

Practice Q.10.3 Verify Euler's theorem for the following function $u = \log\left(\frac{x^2 + y^2}{xy}\right)$

Practice Q.10.4 Verify Euler's theorem for the following function $u = x^2 \log\left(\frac{y}{x}\right)$

Practice Q.10.5 Verify Euler's theorem for the following function $u = \log\left(\frac{x^4 + y^4}{x + y}\right)$

Practice Q.10.6 If $u = x^3 + y^3 + z^3 + 3xyz$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 3u$.

Practice Q.10.7 If $u = \sin^{-1}\left(\frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}}\right)$ then evaluate $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

Practice Q.10.8 If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

Practice Q.10.9 If $u = \sin^{-1}\left(\frac{x^{1/3} + y^{1/3}}{x^{1/2} - y^{1/2}}\right)^{1/2}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{12}\tan u$.


Thank
you

UNIT - 2 LEC - 11

Total Differentiation

(Differentiation of composite functions)

(i) Ordinary Differentiation
(ii) Partial Differentiation
(iii) Total Differentiation

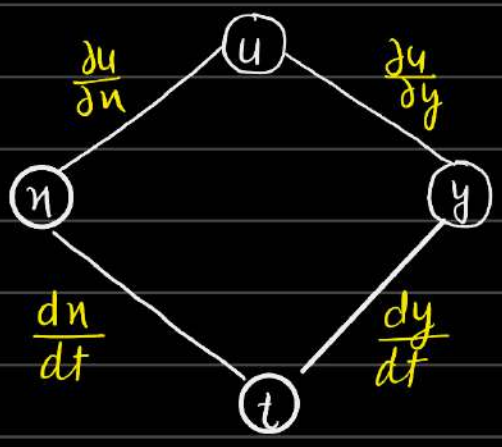
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CASE - I

If u is a composite function of t, such that

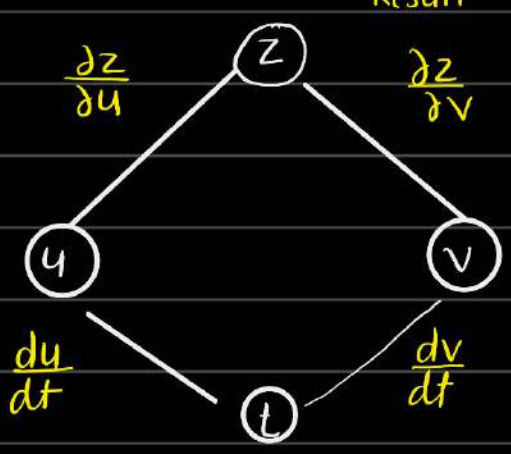
$u = f(x, y)$
$x = \phi_1(t)$
$y = \phi_2(t)$

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt}$$



Q1 - If $z = u^2 + v^2$ and $u = at^2$, $v = 2at$ find $\frac{dz}{dt}$ and verify the Result

$$\frac{dz}{dt} = \frac{\partial z}{\partial u}\cdot\frac{du}{dt} + \frac{\partial z}{\partial v}\cdot\frac{dv}{dt}$$



$$\frac{dz}{dt} = 2u \times 2at + 2v \times 2a$$

$$\frac{dz}{dt} = 4uat + 4va$$

$$\frac{dz}{dt} = 4 \times at^2 \times at + 4 \times 2at \times a$$

$$\frac{dz}{dt} = 4a^2t^3 + 8a^2t \quad \text{—— (1)}$$

2nd Method

$$z = u^2 + v^2$$

$$z = (at^2)^2 + (2at)^2$$

$$z = a^2t^4 + 4a^2t^2$$

diff. w.r.t to t

$$\frac{dz}{dt} = 4a^2t^3 + 8a^2t \quad \text{— (2)}$$

From equation (1) and (2)
Result is verified

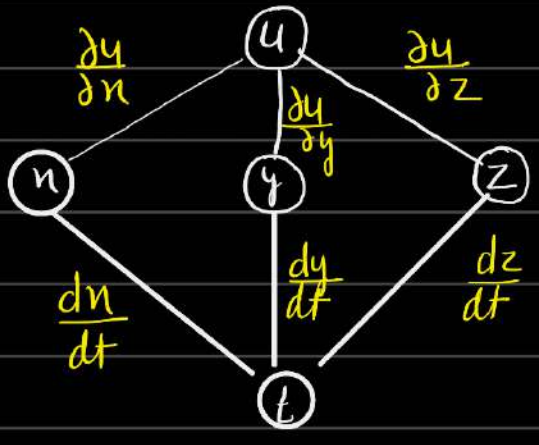
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CASE-2

If u is a composite function of t, such that

$$u = f(x, y, z)$$
$$x = \phi_1(t)$$
$$y = \phi_2(t)$$
$$z = \phi_3(t)$$

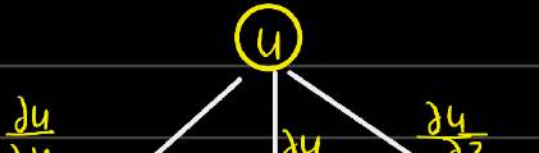


$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt}$$

Q2- Find $\frac{du}{dt}$ as a total Derivative and verify the result by Direct Substitution if $u = x^2 + y^2 + z^2$ and $x = e^{2t}$, $y = e^{2t}\cos 3t$, $z = e^{2t}\sin 3t$

Sol.

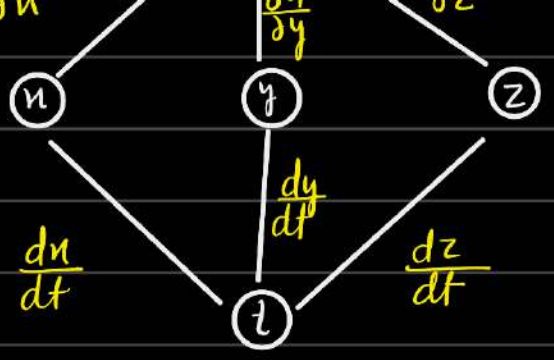
$$u = x^2 + y^2 + z^2$$



$x = e^{2t}$

$y = e^{2t}\cos 3t$

$z = e^{2t}\sin 3t$



$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt}$$

$$\frac{du}{dt} = 2x \times e^{2t} \times 2 + 2y\left[e^{2t}(-\sin 3t)\times 3 + \cos 3t \times e^{2t} \times 2\right] + 2z\left[e^{2t} \times \cos 3t \times 3 + \sin 3t \times e^{2t} \times 2\right]$$

$$\frac{du}{dt} = 4x \times x + 2y(-3e^{2t}\sin 3t + 2e^{2t}\cos 3t) + 2z(3e^{2t}\cos 3t + 2e^{2t}\sin 3t)$$

$$\frac{du}{dt} = 4x^2 + 2y(-3z + 2y) + 2z(3y + 2z)$$

$$\frac{du}{dt} = 4x^2 - 6yz + 4y^2 + 6yz + 4z^2$$

$$\frac{du}{dt} = 4x^2 + 4y^2 + 4z^2$$

$$\frac{du}{dt} = 4(e^{2t})^2 + 4(e^{2t}\cos 3t)^2 + 4(e^{2t}\sin 3t)^2$$

$$\frac{du}{dt} = 4e^{4t} + 4e^{4t}\cos^2 3t + 4e^{4t}\sin^2 3t$$

$$\frac{du}{dt} = 4e^{4t} + 4e^{4t}(\cos^2 3t + \sin^2 3t)$$

$$\frac{du}{dt} = 4e^{4t} + 4e^{4t} = 8e^{4t}$$

$$\boxed{\frac{du}{dt} = 8e^{4t}} \quad \text{— (1)}$$

Also, $u = x^2 + y^2 + z^2$

$u = (e^{2t})^2 + (e^{2t}\cos 3t)^2 + (e^{2t}\sin 3t)^2$

diff w.r.t t

$$u = e^{4t} + e^{4t}\cos^2 3t + e^{4t}\sin^2 3t$$

$u = e^{4t} + e^{4t}(\cos^2 3t + \sin^2 3t)$

$u = e^{4t} + e^{4t}$

$u = 2e^{4t}$

$$\frac{du}{dt} = 8e^{4t} \quad -(2)$$

From equation (1) and (2) Result is verified

UNIT -2

Total Differentiation

LEC-12

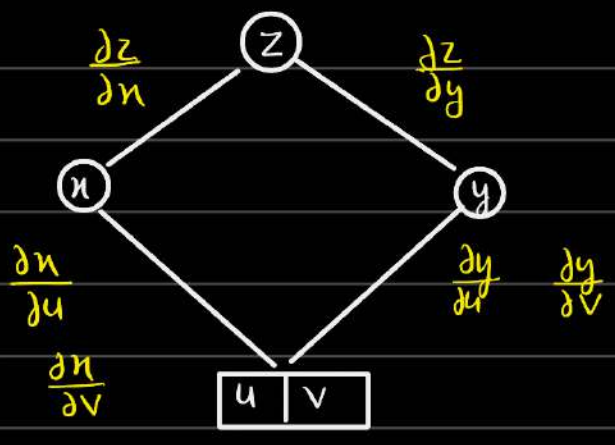
CASE-III

If z is a composite function of u and v such that

$z = f(x, y)$

$x = \phi_1(u, v)$

$y = \phi_2(u, v)$



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

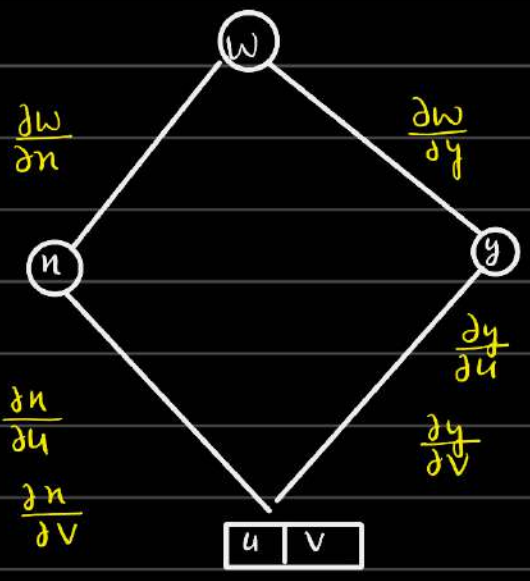
Q1- If $W = f(x, y)$ where $x = e^u \cos v$ and $y = e^u \sin v$ show that

$$y\frac{\partial W}{\partial u} + x\frac{\partial W}{\partial v} = e^{2u}\frac{\partial W}{\partial y}$$

Sol $W = f(x, y)$

$x = e^u \cos v$

$y = e^u \sin v$



$$\frac{\partial W}{\partial u} = \frac{\partial W}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial W}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial W}{\partial u} = \frac{\partial W}{\partial x} \times e^u \cos v + \frac{\partial W}{\partial y} \times e^u \sin v$$

$$\frac{\partial w}{\partial u} = x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y}$$

$$y\frac{\partial w}{\partial u} = xy\frac{\partial w}{\partial x} + y^2\frac{\partial w}{\partial y} \quad \text{---} \quad (1)$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\cdot e^u(-\sin v) + \frac{\partial w}{\partial y}\, e^u \times \cos v$$

$$\frac{\partial w}{\partial v} = -y\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial y}$$

$$x\frac{\partial w}{\partial v} = -xy\frac{\partial w}{\partial x} + x^2\frac{\partial w}{\partial y} \quad \text{---} \quad (2)$$

Adding (1) and (2)

$$y\frac{\partial w}{\partial u} + x\frac{\partial w}{\partial v} = xy\frac{\partial w}{\partial x} + y^2\frac{\partial w}{\partial y} - xy\frac{\partial w}{\partial x} + x^2\frac{\partial w}{\partial y}$$

$$= (x^2+y^2)\frac{\partial w}{\partial y}$$

$$= \left[(e^u\cos v)^2 + (e^u\sin v)^2\right]\frac{\partial w}{\partial y}$$

$$= \left(e^{2u}\cos^2 v + e^{2u}\sin^2 v\right)\frac{\partial w}{\partial y}$$

$$= e^{2u}\left(\sin^2 v + \cos^2 v\right)\frac{\partial w}{\partial y}$$

$$\boxed{y\frac{\partial w}{\partial u} + x\frac{\partial w}{\partial v} = e^{2u}\frac{\partial w}{\partial y}}$$

Hence proved

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UNIT - 2

Total Differentiation

Lec-13

CASE - IV

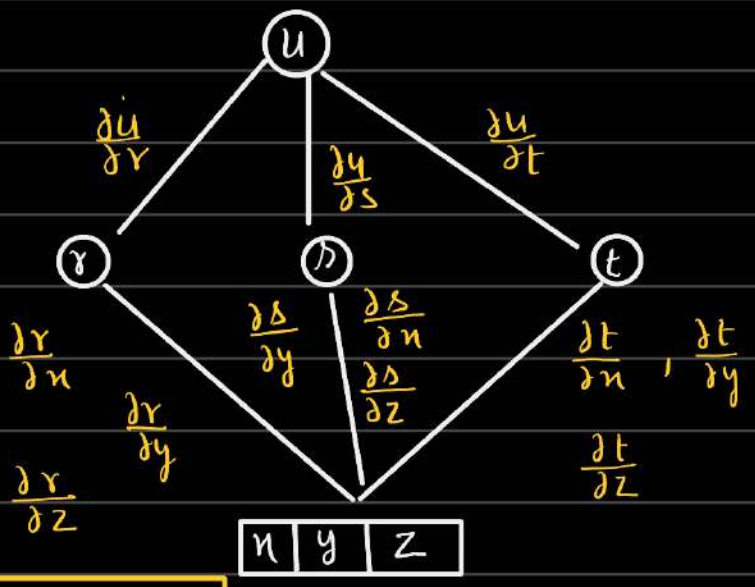
If u is a composite function of x, y, z such that

$u = f(r, s, t)$

$r = \phi_1(x, y, z)$

$s = \phi_2(x, y, z)$

$t = \phi_3(x, y, z)$



$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial z}$$

Q1- If $u = f(y-z, z-x, x-y)$ Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Sol

Let $u = f(r, s, t)$

where

$$r = y - z$$
$$s = z - x$$
$$t = x - y$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot (0) + \frac{\partial u}{\partial s} \cdot (-1) + \frac{\partial u}{\partial t} (1)$$

$$\frac{\partial u}{\partial x} = 0 - \frac{\partial u}{\partial s} + \frac{\partial u}{\partial t}$$

$$\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \text{—— (1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot 1 + \frac{\partial u}{\partial s} \cdot (0) + \frac{\partial u}{\partial t} \cdot (-1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \quad \text{—— (2)}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} (-1) + \frac{\partial u}{\partial s} \cdot (1) + \frac{\partial u}{\partial t} \cdot (0)$$

$$\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \text{—— (3)}$$

Adding equations (1), (2) and (3)

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = -\cancel{\frac{\partial u}{\partial s}} + \cancel{\frac{\partial u}{\partial t}} + \cancel{\frac{\partial u}{\partial r}} - \cancel{\frac{\partial u}{\partial t}} - \cancel{\frac{\partial u}{\partial r}} + \cancel{\frac{\partial u}{\partial s}}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

Hence, proved

AKTU

Engineering Mathematics -1(BAS-103)

Unit-2

Differential Calculus-I

Lec-14

Total Differentiation
(Composite Functions)

As per New Syllabus 2022-23

Case-IV *If u is a composite function of x, y and z, such that*

$$u = f(r,s,t)$$

$$r = \emptyset_1(x,y,z)$$

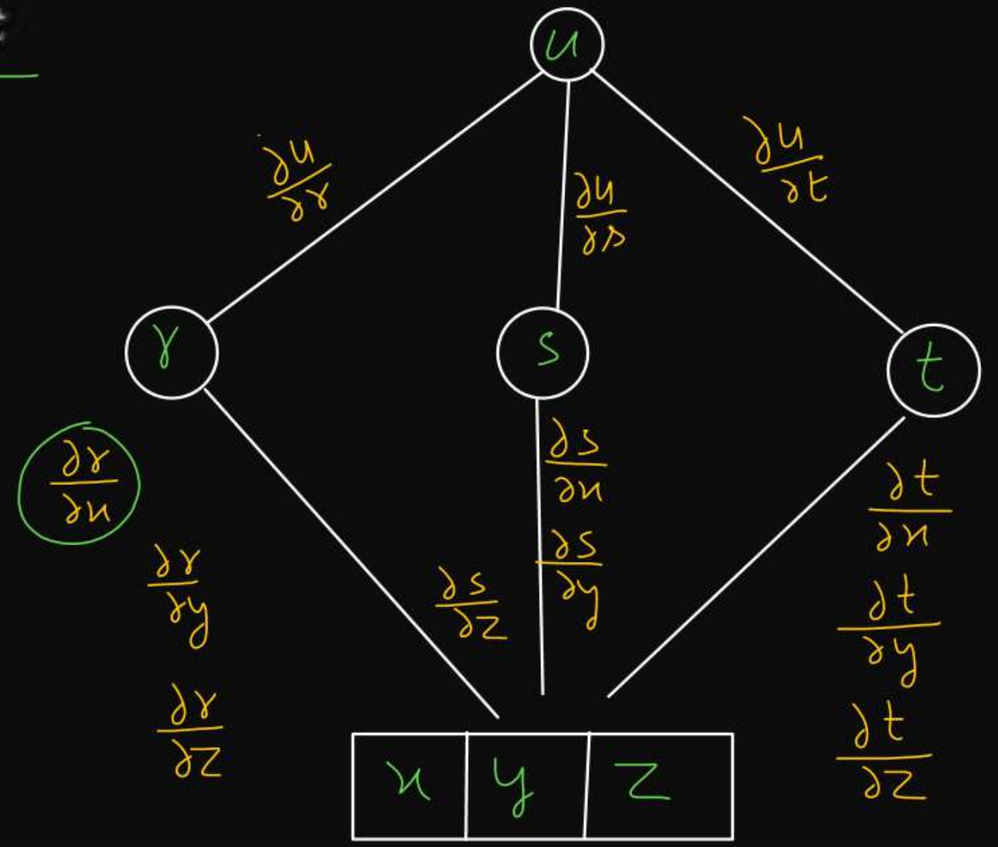
$$s = \emptyset_2(x,y,z)$$

$$t = \emptyset_3(x,y,z)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial z}$$



Q.14.1 If $u = f(y-z, z-x, x-y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. [U.K.T.U. 2010]

Let $u = f(r, s, t)$

Where

$r = y - z$

$s = z - x$

$t = x - y$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial x}$$

$$= \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}(-1) + \frac{\partial u}{\partial t}(1)$$

$$\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \text{—— (1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial y}$$

$$= \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t}(-1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \quad \text{—— (2)}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}(-1) + \frac{\partial u}{\partial s}(1) + \frac{\partial u}{\partial t}(0)$$

$$\frac{\partial u}{\partial z} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \text{—— (3)}$$

Adding equation (1), (2) and (3)

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} + \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} - \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$$

$$\boxed{\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0}$$

Q.14.2 If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}, t = \frac{z}{x}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$. (A.K.T.U. 2018)

$u = f(r, s, t)$

where

$r = \frac{x}{y}$

$s = \frac{y}{z}$

$t = \frac{z}{x}$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \times \frac{1}{y} + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t}\left(\frac{-z}{x^2}\right)$$

$$x\frac{\partial u}{\partial x} = \frac{x}{y}\frac{\partial u}{\partial r} - \frac{z}{x}\frac{\partial u}{\partial t} \quad — (1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\left(\frac{\partial r}{\partial y}\right) + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\left(\frac{-x}{y^2}\right) + \frac{\partial u}{\partial s}\left(\frac{1}{z}\right) + \frac{\partial u}{\partial t}(0)$$

$$y\frac{\partial u}{\partial y} = -\frac{x}{y}\frac{\partial u}{\partial r} + \frac{y}{z}\frac{\partial u}{\partial s} \quad — (2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial z}$$

$$= \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}\left(\frac{-y}{z^2}\right) + \frac{\partial u}{\partial t}\left(\frac{1}{x}\right)$$

$$z\frac{\partial u}{\partial z} = -\frac{y}{z}\frac{\partial u}{\partial s} + \frac{z}{x}\frac{\partial u}{\partial t} \quad — (3)$$

Adding equation (1), (2) and (3)

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{x}{y}\frac{\partial u}{\partial r} - \frac{z}{x}\frac{\partial u}{\partial t} - \frac{x}{y}\frac{\partial u}{\partial r} + \frac{y}{z}\frac{\partial u}{\partial s} - \frac{y}{z}\frac{\partial u}{\partial s} + \frac{z}{x}\frac{\partial u}{\partial t}$$

$$\boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0}$$

Q.14.3 (a) If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$. (U.P.T.U. 2015)

Let $u = f(r, s, t)$

where

$r = 2x - 3y$

$s = 3y - 4z$

$t = 4z - 2x$

Case-V

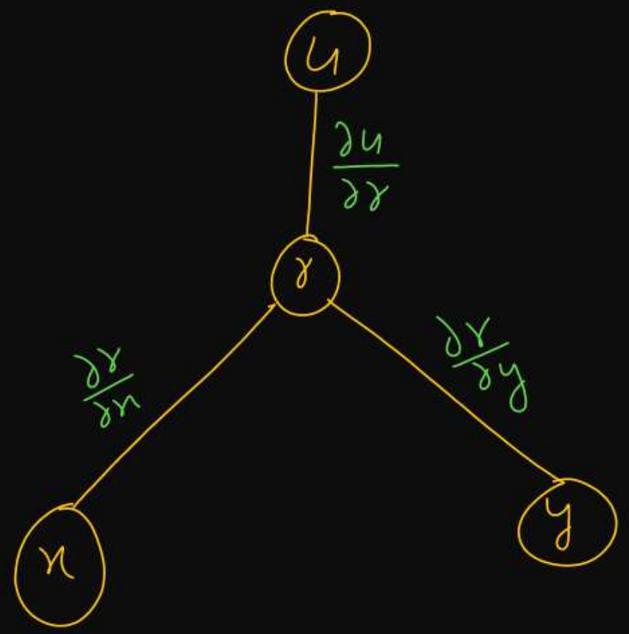
If u is a composite function of x and y such that

u = f(r)

r = Ø(x,y)

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \times \frac{\partial r}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \times \frac{\partial r}{\partial y}$$



Q.14.4 *If $u = f(r)$ and $x = r\cos\theta, y = r\sin\theta$, prove that* $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

$u = f(r)$

$x = r\cos\theta$ — ①

$y = r\sin\theta$ — ②

Squaring and adding ① and ②

$x^2 + y^2 = r^2\cos^2\theta + r^2\sin^2\theta$

$x^2 + y^2 = r^2(\sin^2\theta + \cos^2\theta)$

$$\boxed{r^2 = x^2 + y^2}$$

$2r\frac{\partial r}{\partial x} = 2x$ | $2r\frac{\partial r}{\partial y} = 2y$

$\frac{\partial r}{\partial x} = \frac{x}{r}$ | $\frac{\partial r}{\partial y} = \frac{y}{r}$

$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x}$

$\frac{\partial u}{\partial x} = f'(r) \times \frac{x}{r}$

$\frac{\partial u}{\partial x} = \left(\frac{x}{r}\right) f'(r)$

Partially Diff. w.r.t x

$\frac{\partial^2 u}{\partial x^2} = \frac{x}{r} f''(r) \times \left(\frac{\partial r}{\partial x}\right) + f'(r)\left[\frac{r \times 1 - x\frac{\partial r}{\partial x}}{r^2}\right]$

$\frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2} f''(r) + \frac{1}{r^2} f'(r)\left[r - x \times \frac{x}{r}\right]$

$= \frac{x^2}{r^2} f''(r) + \frac{1}{r^2} f'(r)\left(\frac{r^2 - x^2}{r}\right)$

$\frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2} f''(r) + \frac{1}{r^2} f'(r) \times \frac{y^2}{r}$

$$\boxed{\frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2} f''(r) + \frac{y^2}{r^3} f'(r)} \quad \text{— ①}$$

$$\boxed{\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{r^2} f''(r) + \frac{x^2}{r^3} f'(r)}$$

2 — ②

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{x^2}{r^2} f''(r) + \frac{y^2}{r^3} f'(r) + \frac{y^2}{r^2} f''(r) + \frac{x^2}{r^3} f'(r)$$

$$= \left(\frac{x^2}{r^2} + \frac{y^2}{r^2}\right) f''(r) + \left(\frac{y^2}{r^3} + \frac{x^2}{r^3}\right) f'(r)$$

$$= \left(\frac{x^2 + y^2}{r^2}\right) f''(r) + \left(\frac{x^2 + y^2}{r^3}\right) f'(r)$$

$$= \left(\frac{\cancel{r^2}}{\cancel{r^2}}\right) f''(r) + \left(\frac{\cancel{r^2}}{r^{3-2}}\right) f'(r)$$

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)}$$


Thank
you